

Project Thesis
PRO-089



Roll Control and Estimation of a Canard-controlled Sounding Rocket

Rollregelung und Schätzung einer
Canard-gesteuerten Höhenforschungsrakete

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Contents

1	Introduction	1
1.1	Motivation	1
1.2	Problem statement	2
1.3	Thesis structure	2
2	Background	3
2.1	Sounding rocket flight dynamics	3
2.2	Guidance, navigation, and control	6
2.3	Demonstrator system architecture	8
3	Model	11
3.1	Kinematics & dynamics	11
3.2	Sensor measurements	14
3.3	Atmosphere and air data	15
3.4	Aerodynamic forces and torques	17
3.5	Actuator model	19
4	Controller	21
4.1	Optimal control	22
4.2	Reduced controller model	23
4.3	Scheduled control law	24
4.4	Response analysis	26
5	Estimator	29
5.1	Optimal estimation	29
5.2	Estimator models	31

5.3	EKF algorithm	33
5.4	Pad filter	35
6	Simulation	37
6.1	ClosedRocket simulation framework	37
6.2	Simulation results	39
6.2.1	Aurora nominal	39
6.2.2	Coning analysis	41
6.2.3	Expected flight attitude	43
6.3	Monte-Carlo simulations	44
6.4	Hardware-in-the-loop	47
7	Flight results	51
7.1	Testflight results	51
7.2	Aurora flight results	53
8	Conclusion	57
8.1	Summary	57
8.2	Future work	58
	Bibliography	59
A	Contents Archive	65
B	Jacobians	67
C	Further models	73

Chapter 1

Introduction

1.1 Motivation

Sounding rockets are suborbital research vehicles that can reach altitudes anywhere from 10 km to over 1500 km [BurthEtAl23]. They are usually launched by space agencies to conduct atmospheric and other research, such as the ESA TEXUS missions [SchütteGrothe05] or the NASA Sounding Rocket Program [Adkins25]. Additionally, they are increasingly developed by university teams for educational purposes and competitions, such as the European Rocketry Challenge [EuRoC25], Launch Canada [LC25], NASA student launch [NASA25], or IREC [ESRA25]. Sounding rockets undergo rapidly changing dynamics, often with accelerations exceeding 10 g and with velocities of over 500 m/s producing high aerodynamic forces.

Aerodynamic instabilities can result in trajectory deviations and excessive structural loading, which may cause harm to people and damage or destroy the vehicle. Maintaining a stable flight attitude is thus critical for mission success and safety. Most sounding rockets are passively stabilized in the pitch and yaw directions by tail fins [Siouris04, BabbFuller67]. However, rockets with this stabilization method still experience large roll motions, and coupling of roll and pitch dynamics can result in a self-amplifying precession called roll lock-in [Lawrence65]. Active attitude control, particularly of the roll axis, is therefore often desired for high-performance missions.

In launchers and missiles, a Guidance, Navigation, and Control (GNC) system ensures a stable and accurate flight by providing attitude and trajectory determination, stabilization, and tracking. Many sounding rocket missions also feature a GNC system which provides some form of attitude control, often using canards [BurthEtAl23, Ljunge13]. Sounding rockets have only limited ground communication and must rely on onboard systems for autonomous navigation and control. Reliable state estimation and control laws are therefore essential.

1.2 Problem statement

High-altitude sounding rockets are prone to roll lock-in (Sect. 2.1). At the University of Waterloo, sounding rockets are developed by the Waterloo Rocketry student team (Sect. 2.3). Aiming for higher altitudes, the lock-in phenomenon becomes increasingly a concern for flight stability. A canard control system is developed by the team which serves to reduce the roll motion and thus mitigate the chance of lock-in. Additionally, accurate determination of the rocket's flight data is desired.

The aim of this thesis is therefore to develop a controller for regulating the rocket's roll rate by actuating the canards, and to develop a state estimator for attitude and trajectory determination in flight.

A central challenge are the rapidly changing dynamics of sounding rocket flight, the controller must account for these varying flight conditions. Furthermore, determining the exact aerodynamic behavior of the canards and their interaction with the rocket is difficult. Therefore the controller and estimator must be robust against model uncertainty, especially with respect to the canards.

1.3 Thesis structure

The thesis is structured as follows. Chap. 2 presents a review of existing literature, including sounding rocket flight dynamics and active control methods. Furthermore, it also introduces the demonstrator system. The following chapters discuss the methodology: In Chap. 3 the rocket flight is modeled, with a focus on the rigid-body dynamics as well as the aerodynamics and measurement models required for control and estimation design. The LQR-based controller is designed in Chap. 4, and the EKF-based estimator is developed in Chap. 5. The results are discussed in the next two chapters: Simulation and analysis of the closed-loop behavior is presented in Chap. 6. In Chap. 7 the resulting data from two flights is analyzed. Finally, Chap. 8 holds concluding remarks.

Chapter 2

Background

In this chapter related research is presented. This includes a background on rocket flight dynamics and the roll lock-in phenomenon in Sect. 2.1, and research into flight control and estimation in Sect. 2.2. The demonstrator sounding rocket Aurora with the canard control system is introduced in Sect. 2.3.

2.1 Sounding rocket flight dynamics

The aerodynamic stability of slender-body rockets has been studied for decades [Barrowman67, BabbFuller67]. A rocket is considered statically stable if its center of pressure is located aft of the center of gravity, as shown in Fig. 2.1. Any deviation from straight flight or wind disturbances cause a restoring moment, turning the rocket back into alignment with the velocity vector. This response is stable, but weakly damped and thus pitch oscillations occur. Larger tail fins move the center of pressure further aft, increasing restoring and damping moments and thereby improving static stability. The aerodynamic coefficients of slender rockets vary significantly with flight conditions, especially at high altitudes and Mach numbers [BabbFuller67]. Barrowman developed a method for predicting the aerodynamic characteristics of rockets from geometry. Wind tunnel and flight tests validated the predictions for both subsonic and supersonic flight regimes [Barrowman67]. This method is still used for stability determination and design in sounding rocket development [Niskanen09, RogersCooper19].

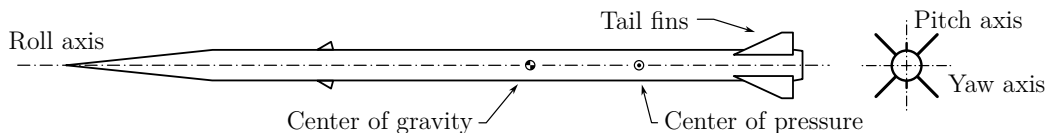


Figure 2.1: Sounding rocket geometry.

Roll motion of the rocket can lead to coning, an oscillatory motion in which the longitudinal axis precesses about the velocity vector [Siouris04]. Coning typically arises when a rolling rocket is subjected to a disturbance in pitch or yaw, e.g. from wind gusts, propellant slosh, or roll-induced moments from the Magnus effect [Platus98]. Small asymmetries within the vehicle, such as thrust misalignment, unequal fins, or off-axis mass distribution can also create a constant trim disturbance in either axis [Lawrence65]. For most roll rates any pitch disturbance causes a stable coning motion, and at low amplitudes this is an often benign characteristic of stable flight. However, rockets rolling with a rate similar to their natural pitch frequency are prone to a coning condition known as roll lock-in, especially at high velocities.

Roll lock-in problem

When the ratio of roll rate and natural pitch frequency approaches one, the roll and pitch motions phase-lock [Lawrence65]. This coupling leads to resonance, resulting in large oscillations where the airstream always hits the same side of the rocket [Brunk66]. Self-amplification at resonance produces an unstable coning motion with a runaway angle-of-attack, this can happen quickly under high dynamic pressure [Lawrence65, Platus98]. Aerodynamic loads rise rapidly for unstable angles-of-attack, and can cause loss of control, structural failure, and vehicle destruction within seconds. Even in non-catastrophic cases, roll lock-in reduces the apogee, risks unsuccessful recovery, and large trajectory deviations pose a risk to the public. The aerodynamic moments acting on the rocket are generally nonlinear and depend on the instantaneous angle-of-attack and roll rate, which makes the motion chaotic. As the angle-of-attack increases, induced aerodynamic forces can grow stronger further amplifying the coning motion [HatafskySollowTaylor67, Clare71]. Depending on the specific asymmetries and flight conditions, the rocket may exhibit stable small-amplitude oscillations, large erratic oscillations, or diverging motion [AnanthkrishnanRaisinghani92]. This sensitivity makes roll lock-in difficult to predict or reproduce consistently across flights, and therefore complete prevention of roll lock-in through design alone is difficult to achieve. Sufficient static stability margins and minimal asymmetries are good design practices to reduce the likelihood of resonance, but they do not guarantee complete mitigation [HatafskySollowTaylor67, HoultTran15]. A roll rate either much lower or much higher than the natural pitch frequency avoids the resonance condition [Brunk66, HoultTran15]. This is often utilized in spin-stabilized rockets, rolling much faster than their pitch frequency. However, these vehicles pass through the resonance frequency at some point, risking lock-in which the vehicle may or may not escape [Lawrence65, HatafskySollowTaylor67]. In contrast, active roll control systems can maintain a very low or zero roll rate, and robust feedback control suppresses otherwise destabilizing disturbances.

Aerodynamic control surfaces

Control surfaces, such as canards or fin tabs shown in Fig. 2.2, offer high control authority during high-speed atmospheric flight due to aerodynamic forces.

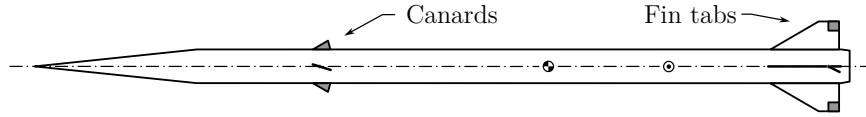


Figure 2.2: Rocket control surfaces: canards and fin tabs.

Fin tabs are control surfaces located at the trailing edge of a tail fin, similar to ailerons on airplanes. They are either driven by a motor or are purely mechanical systems such as rollerons, roll dampers used in early guided missiles. Rollerons consist of a rapidly spinning disk embedded in the freely hinged fin tab. Under roll motion of the missile the gyroscopic precession of the disk deflects the tab into the airstream, counteracting the roll. However, if not properly tuned, rollerons can introduce roll rate oscillations with limit cycle behavior [BrownNason54].

Canards are control surfaces ahead of the center of gravity [Siouris04]. Compared to fin tabs their location allows for easier integration into the vehicle, as the space further aft is confined due to propulsion systems. However, the aerodynamic interaction between canards and tail fins can be complex, as canards produce shock waves at supersonic speeds and actuation sheds vortices as illustrated in Fig. 2.3. Blair conducted a study of canard-configured missiles in supersonic wind tunnel conditions. Canard-induced vortices lead to an effective reversal of the roll torque in interaction with fixed tail fins [Blair78]. Configurations where the tail fin assembly could freely spin around the roll axis did not exhibit this roll reversal, such that the missile could be controlled successfully in roll with conventional methods [Blair78]. Both configurations had similar longitudinal static stability [Blair78]. Landers examined aerodynamic predictions for various canard-controlled missile geometries. The aerodynamic coefficients, especially for the roll torques, are sensitive to the specific geometry of the canard-fin arrangement [LandersHallAuman00]. While high-fidelity aeroprediction codes for canard-fin interactions were validated by wind tunnel data, these are for military use and are not publicly accessible. Lesieutre also showed that freely rolling tail fins reduced susceptibility to roll reversal when using canards [LesieutreQuijano14].

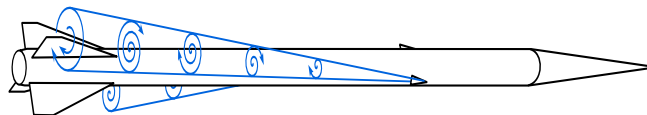


Figure 2.3: Canard-induced vortices interacting with the tail fins.

2.2 Guidance, navigation, and control

This section treats previous work in sounding rocket control and estimation, with a focus on the canard-based control systems. The primary objective of a GNC system in sounding rockets is to reduce trajectory dispersion and to maintain the desired attitude throughout the flight, thereby improving vehicle recovery and payload pointing quality. Active guidance can compensate early-flight perturbations such as thrust misalignment, roll spin-up, and wind cocking where the rocket turns into the horizontal ground-level wind. Guided rockets have been shown to achieve significantly higher trajectory repeatability and pointing accuracy than unguided counterparts [Ljunge13].

Beyond Gravity (formerly RUAG Space) developed the S19 family of canard-based guidance systems, shown in Fig. 2.4a, to provide attitude stabilization after launch [Ljunge13]. Notable is NASA’s use of the S19L system in its sounding rocket program [BurthEtAl23]. The system provides control in pitch and yaw while, maintaining a nominal spin rate of approximately 4 Hz, well above the natural pitch frequency, to avoid roll lock-in. Prior to launch the system determines the rocket’s initial attitude, which serves as control reference during ascent.

Canard-based GNC systems have also been developed within rocketry student teams. DARE, the TU Delft rocketry team, designed and flew a canard roll control system. Wind tunnel tests, as shown in Fig. 2.4b, showed successful pitch control, and flight tests demonstrated adequate roll control although they revealed roll reversal effects [DARE25]. UC Rocketry from the University of Canterbury implemented a roll damping and yaw control system for a small rocket. A flight test demonstrated adequate yaw and roll stabilization, but during a yaw step the axes coupling led to a temporary loss of roll control, which could have been due to interference with the fixed tail fins [BuchananEtAl15]. The Uni-



(a) BeyondGravity S19E [BeyondGravity25].



(b) DARE VS7-ACT [DARE25]

Figure 2.4: Canard-based control systems.

versity of Leeds rocketry team developed a canard attitude stabilization system to mitigate pitch deviations caused by wind cocking, using a linear quadratic regulator (LQR). Although the small rocket had freely rotating fins for roll reversal mitigation, which also should have reduced coning effects, flight data still showed a significant precession of the longitudinal axis [PostaEtAl24]. This could be due to inadequate tuning or lacking robustness in the control loop, causing oscillations.

During rocket flight the flight conditions, such as Mach number and dynamic pressure, undergo large variations. Classical linear control techniques often reach their limits in these cases, so gain-scheduled and nonlinear control laws have been developed to accommodate these variations. Buschek implemented a gain-scheduled robust control law for a thrust-vector-controlled missile, formulating the lateral dynamics as a linear parameter-varying (LPV) system. The control gains were interpolated across dynamic pressure regimes, yielding good tracking performance [Buschek99]. Song developed quaternion-based robust control. The method demonstrated good tracking and stability in simulation, but required fully controllable attitude dynamics [SongEtAl01]. Fresconi developed an LQR roll controller for a canard-stabilized artillery projectile, where actuator dynamics were included in the model. The controller successfully regulated roll motion in both simulation and wind tunnel testing [FresconiEtAl14].

The extended Kalman filter (EKF) has become a standard tool for state estimation in aerospace navigation [LewisXiePopa08]. Minh used an EKF for post-flight state estimation of DARE’s sounding rockets. The EKF performed better than an Unscented Kalman Filter in both accuracy and computational effort [Minh12]. Additionally, modeled-force and accelerometer-input approaches were compared, which showed that accelerometer-input provided higher accuracy compared to models with force formulations [Minh12]. MacLean investigated closed-loop control simulations for sounding rocket with EKF-based state estimation. However, inadequate fidelity limited the accuracy of the results, and in subsequent flight tests only partially valid estimation data was produced [MacLean17].

Quaternion-based dynamics are an effective formulation for attitude determination, avoiding the singularities and discontinuities of Euler-angle kinematics. Extended Kalman filters with quaternion models are often used [Solà17], and variations of specific filter formulations are an ongoing area of research [BonnabelMartinSalaun09, MayhewSanfeliceTeel11, KangEtAl13, BourmaudEtAl13, SveierSjøbergEgeland19].

2.3 Demonstrator system architecture

This thesis focuses on Aurora, a liquid-bipropellant sounding rocket under development by the Waterloo Rocketry student team [WR25]. This team is representing the University of Waterloo in the 2025 Launch Canada Challenge [LC25]. Aurora is 5.1 m long with a diameter of 20.3 cm, has a dry mass of 60 kg and carries roughly 50 kg of propellants. The liquid bi-propellant blowdown engine uses nitrous oxide as its oxidizer and ethanol as fuel. The engine burns for about 10 s with a maximum thrust of roughly 10 kN. During its flight at Launch Canada, shown in Fig. 2.5, Aurora reached a velocity over Mach 2 and an apogee of 12 km above sea level. Next to the engine components and a parachute recovery system, Aurora carries a canard control system for roll regulation.



Figure 2.5: Aurora lift-off at Launch Canada [WR25].

Aurora uses two canards to generate pure rolling moments on the vehicle. Canards were chosen over fin tabs as the actuation system was easier to integrate forward of the propellant tanks rather than aft, where available space next to the propulsion systems is confined. Aurora has a freely rolling tail fin assembly to reduce the susceptibility to vortex-induced roll reversal, this also serves to further reduce coning motion. The values of rocket parameters used in the model (Chap. 3) are shown in Tab. 2.1.

Table 2.1: Modeling parameters used for Aurora. Mass and inertia are for the empty rocket, distances are relative to the center of gravity.

Symbol	Value	Unit	Description
m	54.9	kg	Rocket mass, empty
J_{xx}	0.46	kg m ²	Roll moment of inertia, empty
J_{yy}	49.5	kg m ²	Pitch/yaw moment of inertia, empty
d_{cp}	-0.5	m	Overall center of pressure, axial
A_{ref}	0.32	m ²	Overall reference area
C_{N_α}	10		Overall normal force coefficient deriv.
d_{can}	0.12	m	Canard center of pressure, radial
A_{can}	-0.0052	m ²	Canards total planform area
Λ_{can}	61	deg	Canard sweep angle
$\mathbf{d}_{a,1}^\top$	[1.2, 0.074, -0.027]	m	Accelerometer mounting location
$\mathbf{d}_{a,2}^\top$	[1.2, 0.065, 0.047]	m	Accelerometer mounting location

Canard control system

The canard shape was designed to exhibit low drag and a subsonic leading edge, and the size is chosen such that the rocket's static stability margin is not significantly reduced [Scavone25]. To prevent stall of the control surfaces at supersonic speeds, the canards are mechanically limited to actuate by a maximum of $\pm 10^\circ$. To provide only roll torques on the rocket, the mechanical structure was designed to ensure equal actuation of the canards [Pickens25]. A servo motor drives the canard shafts via bevel gears, as shown in Fig. 2.6a, rotating the two canards by equal amounts in opposite directions. For accurate position feedback to the estimator a rotary potentiometer is mounted on one canard shaft, as the servo motor does not output its encoder data. Aurora has multiple onboard sensors in the form of MEMS devices. An overview of the redundant sensor suite, with accelerometers, rate gyroscopes, magnetometers, and barometers, is given in Tab. 2.2. While a GPS may be useful for determining the horizontal position, it is not used for the control system as it is unavailable during most of the ascent due to operational limits of commercially available GPS receivers. The electrical system, shown in Fig. 2.6b, includes the motor control board and the processor board, as well as the two strap-down sensor packages.

The control algorithms and simulation presented in this thesis were developed in MATLAB. For real-time execution on the STM32H7 processor the firmware is realized in C [LiEtAl25]. The control algorithms were manually translated from MATLAB to C, which was preferred over using MATLAB CODER as this allowed for better control over specific matrix algorithms and memory management. Extensive unit testing was performed to validate the C implementation

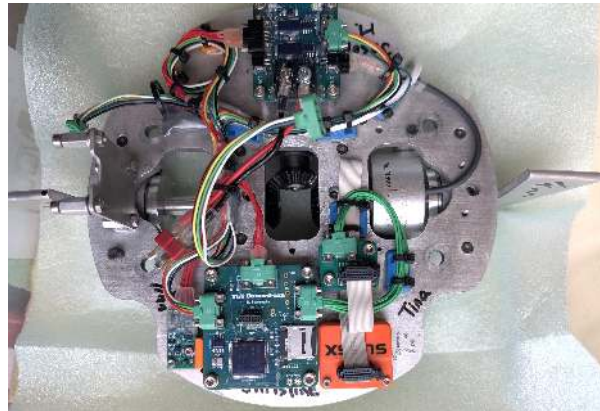
of the controller and estimator functions, including all model functions. Additionally, the processor was subjected to hardware-in-the-loop tests to validate the entire control loop code. Further details on implementation and testing of the firmware exceed the scope of this thesis.

Table 2.2: Onboard sensor suite of Aurora. Available sensors are marked X.

Device name	Accel.	Gyro.	Magn.	Baro.
Movella Xsens MTi-630	X	X	X	X
STM LSM6DSO	X	X	-	-
STM LIS3MDL	X	-	X	-
STM LPS22DF	-	-	-	X



(a) Canards mounted on the rocket. Servo motor, rotary potentiometer, canards, gear train, camera, battery box.



(b) Electrical harness. Motor control board, processor board, sensors, wiring harness.

Figure 2.6: The Canard Control System of Aurora.

Chapter 3

Model

In this chapter the rocket flight is modeled. Dynamical models are needed for both the controller synthesis and the estimator algorithm, and the latter also requires models of the measurement processes.

3.1 Kinematics & dynamics

This section treats the kinematics and dynamics of the rigid body motion of a sounding rocket, the external forces and torques acting on the vehicle are derived in Sect. 3.4. The rocket is assumed to be a single rigid body. The attitude of the rocket is expressed as quaternions, the formulations are given in this section.

Coordinate frames

The body-fixed frame B is aligned along the main inertia axes of the rocket body, as shown in Fig. 3.1. These axes are oriented forward-starboard-rail, i.e. x^B points forward, and y^B is horizontal before launch. The origin of the body-fixed frame is located at the rocket's center of gravity. The center of gravity moves during the boost phase due to changing propellant mass, but a constant location of the dry center of gravity is assumed for now.

The geographic frame G is assumed to be on the local-level plane tangent to Earth's surface, as shown in Fig. 3.2. The orientation is upward-crossrange-downrange, i.e. x^G points upward, and z^G points horizontally in the direction of launch. The origin is located below the launch rail with zero altitude at sea level. The rotation and orbital motion of the Earth can be neglected for the short time interval of the rocket flight [StevensJohnsonLewis15]. Earth's gravity is defined as $\mathbf{g}^G = [-g_0 \ 0 \ 0]^\top$ with a constant magnitude of $g_0 \approx 9.81 \frac{\text{m}}{\text{s}^2}$. As G is an accelerated frame it differs from the inertial frame by the gravity vector $\mathbf{g}^G$.

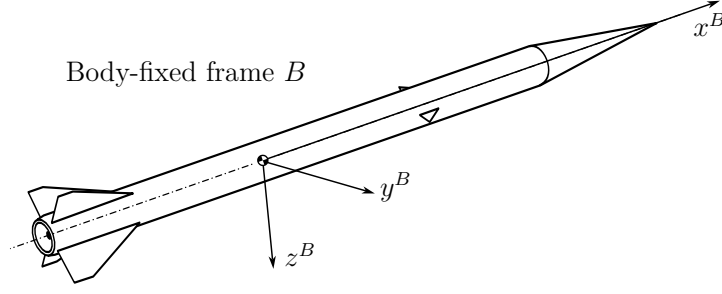


Figure 3.1: Diagram of the rocket body-fixed frame.

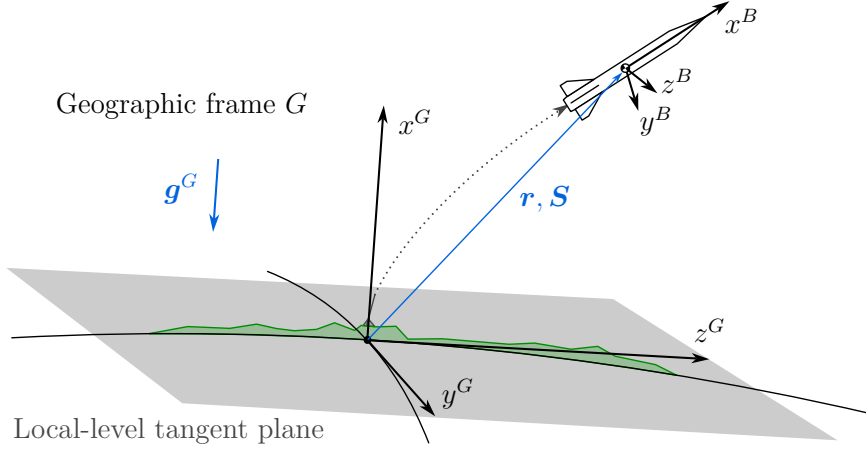


Figure 3.2: Diagram of the Earth-flat geographic frame.

The rotation matrix $\mathbf{S} = \mathbf{S}_G^B$ transforms a vector from the geographic frame G to the body-fixed frame B . The inverse rotation matrix is $\mathbf{S}^{-1} = \mathbf{S}^\top = \mathbf{S}_B^G$. The rotation of a vector $\mathbf{v}$ between the frames is given by

$$\mathbf{v}^B = \mathbf{S} \mathbf{v}^G, \quad \mathbf{v}^G = \mathbf{S}^\top \mathbf{v}^B. \quad (3.1)$$

For simplification of notation the indices G and B are omitted from the following vectors, unless specified otherwise. The position $\mathbf{r}$ of the rocket relative to the origin of G is given in the geographic frame, so

$$\mathbf{r} = \mathbf{r}^G = \begin{bmatrix} r_x & r_y & r_z \end{bmatrix}^\top. \quad (3.2)$$

The velocity $\mathbf{v}$, the angular rates $\boldsymbol{\omega}$, and the external forces $\mathcal{F}$ and torques $\mathcal{T}$ are given in the body-fixed frame, so

$$\mathbf{v} = \mathbf{v}^B = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix}^\top, \quad (3.3)$$

$$\boldsymbol{\omega} = \boldsymbol{\omega}^B = \begin{bmatrix} \omega_x & \omega_y & \omega_z \end{bmatrix}^\top. \quad (3.4)$$

Quaternion attitude

In aeronautics Euler angles are often used as attitude parameters, however the Euler angles kinematics have singularities at certain orientations. Quaternions are used as singularity-free rotation parameters with a linear time derivative. The quaternion is defined as the number $q = q_w + \mathbf{i}q_x + \mathbf{j}q_y + \mathbf{k}q_z$, where all components are orthogonal and $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$. For convenience the quaternion is notated as the vector

$$\mathbf{q} = \begin{bmatrix} q_w & q_x & q_y & q_z \end{bmatrix}^\top = \begin{bmatrix} q_w \\ \mathbf{q}_v \end{bmatrix}, \quad (3.5)$$

with four elements as the real part q_w and the vector part $\mathbf{q}_v$. To describe rotations the quaternion is normed to unity as $\|\mathbf{q}\| = 1$. According to Euler's rotation theorem every rotation in 3D space can be expressed by an angle φ and a unit axis $\mathbf{n}$. The unit quaternion can be understood as that rotation, with

$$q_w = \cos\left(\frac{\varphi}{2}\right), \quad \mathbf{q}_v = \mathbf{n} \sin\left(\frac{\varphi}{2}\right), \quad (3.6)$$

such that zero rotation is expressed as the identity quaternion $q_w = 1$, $\mathbf{q}_v = \mathbf{0}$. Quaternion multiplication uses the Hamiltonian product operation $*$. For the quaternions $\mathbf{p}, \mathbf{q}, \mathbf{q}$ this is [Solà17]

$$\mathbf{p} = \mathbf{q} * \mathbf{q} = \tilde{\mathbf{Q}} \mathbf{q} \quad \text{with} \quad \tilde{\mathbf{Q}} = \mathbf{q} * = \begin{bmatrix} q_w & -q_x & -q_y & -q_z \\ q_x & q_w & -q_z & q_y \\ q_y & q_z & q_w & -q_x \\ q_z & -q_y & q_x & q_w \end{bmatrix}, \quad (3.7)$$

which is not commutative ($\mathbf{q} * \mathbf{q} \neq \mathbf{q} * \mathbf{q}$). A rotation sequence is computed by multiplication. The time derivative of the quaternion attitude with angular rates in the body-fixed frame is the linear system [Zipfel07]

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} * \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}. \quad (3.8)$$

The rotation matrix $\mathbf{S}(\mathbf{q})$ can be computed from the quaternion as [Zipfel07]

$$\mathbf{S}(\mathbf{q}) = \begin{bmatrix} q_w^2 + q_x^2 - q_y^2 - q_z^2 & 2q_wq_z + 2q_xq_y & 2q_xq_z - 2q_wq_y \\ 2q_xq_y - 2q_wq_z & q_w^2 - q_x^2 + q_y^2 - q_z^2 & 2q_wq_x + 2q_yq_z \\ 2q_wq_y + 2q_xq_z & 2q_yq_z - 2q_wq_x & q_w^2 - q_x^2 - q_y^2 + q_z^2 \end{bmatrix}. \quad (3.9)$$

The Euler angles yaw ψ , pitch θ , and roll ϕ can be recovered from the rotation matrix elements. For the rotation sequence yaw, pitch, roll (axes $z \rightarrow y' \rightarrow x''$) the Euler angles are

$$\psi = \text{atan2}(S_{12}, S_{11}), \quad \theta = \arcsin(-S_{13}), \quad \phi = \text{atan2}(S_{23}, S_{33}). \quad (3.10)$$

Relative motion

The position time derivative in the geographic frame is

$$\dot{\mathbf{r}} = \mathbf{S}^\top \mathbf{v}. \quad (3.11)$$

The time derivatives of the body velocity and the body angular rate are formulated using the Newton-Euler equations [Zipfel07, StevensJohnsonLewis15]. By derivation in the rotating body-fixed frame the acceleration is

$$\dot{\mathbf{v}} = \mathbf{a} - \boldsymbol{\omega} \times \mathbf{v} + \mathbf{S}\mathbf{g}^G \quad \text{with} \quad \mathbf{a} = \mathcal{F}/m, \quad (3.12)$$

where $\mathcal{F}$ are the external forces and m is the body mass. The gravity vector $\mathbf{g}^G$ is the acceleration between the inertial and the geographic frame, gravity is thus not an external force. The angular acceleration in the body-fixed frame is

$$\dot{\boldsymbol{\omega}} = \mathbf{J}^{-1}(\boldsymbol{\mathcal{T}} - \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega}), \quad (3.13)$$

which includes the external torques $\boldsymbol{\mathcal{T}}$ and the gyroscopic moment. The inertia matrix $\mathbf{J}$ can be determined from CAD or experimentally, the values derived from CAD are listed in Tab. 2.1. Due to the rotational symmetry of rockets the inertia is typically $\mathbf{J} \approx \text{diag}[J_{xx} \ J_{yy} \ J_{zz}]$, with $J_{xx} < J_{yy} = J_{zz}$ [Siouris04]. Both mass and inertia decrease from liftoff to burnout as propellant is expelled. For the controller design this variation is simplified to be constant with mass and inertia of the empty rocket, as this “worst-case” requires the highest controller bandwidth.

3.2 Sensor measurements

The onboard sensor suite consists of accelerometers, rate gyroscopes, magnetometers, barometers, and the canard shaft encoder. Measurements signals are denoted by an underbar. It is assumed that the axes of each sensor have the same orientation as the body-fixed frame, this is generally not the case but can be corrected before the filter. All sensors are prone to high frequency noise and low frequency biases, among other disturbances. The bias $\mathbf{b}$ is assumed to be constant. The noise $\mathbf{n}$ is assumed to be a white, uncorrelated, zero-mean Gaussian process with variance σ^2 , such that

$$\mathbf{n} \sim \mathcal{N}(0, \boldsymbol{\Sigma}) \quad \text{with} \quad \boldsymbol{\Sigma} = \sigma^2 \mathbf{I}. \quad (3.14)$$

The accelerometer, mounted at the location $\mathbf{d}_a$ relative to the rocket center of gravity, measures the acceleration vector $\underline{\mathbf{a}}$. Including the transformation from the body frame to the sensor frame the measurement is computed as

$$\underline{\mathbf{a}} = \mathbf{a} + \dot{\boldsymbol{\omega}} \times \mathbf{d}_a + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{d}_a) + \mathbf{b}_a + \mathbf{n}_a, \quad (3.15)$$

with the specific force $\mathbf{a}$ in the body-fixed frame. To reduce complexity of the estimator model, the acceleration measurement will be used for the velocity derivative in Eq. (3.12) instead of force modeling [Minh12].

The rate gyroscope measures the angular velocity vector $\underline{\omega}$ as

$$\underline{\omega} = \omega + \mathbf{b}_\omega + \mathbf{n}_\omega. \quad (3.16)$$

The magnetometer measures the local magnetic field vector $\mathbf{m}^B$. This is predominantly Earth's magnetic field $\mathbf{m}_E$, but can include disturbances from local sources on the ground. The field is assumed to be approximately constant over the flight regime, as detailed models exceed the scope of this thesis. The bias elements are mostly the hard iron disturbances $\mathbf{b}_m$ from nearby permanent magnets on the rocket. The measured magnetic field is thus

$$\underline{\mathbf{m}} = \mathbf{S}\mathbf{m}_E^G + \mathbf{b}_m + \mathbf{n}_m. \quad (3.17)$$

The barometer measures the air pressure $\underline{p}$ of the atmosphere (Sect. 3.3) as

$$\underline{p} = p + b_p + n_p. \quad (3.18)$$

The temperature cannot be measured reliably, as the sensor temperature can be much higher than the surrounding air due to the Sun heating up the rocket body.

A rotary potentiometer measures the canard angle as

$$\underline{\delta} = \delta + b_\delta + n_\delta. \quad (3.19)$$

3.3 Atmosphere and air data

The static pressure p , temperature T , density ρ , and the local speed of sound c of the surrounding air vary with altitude, as shown in Fig. 3.3.

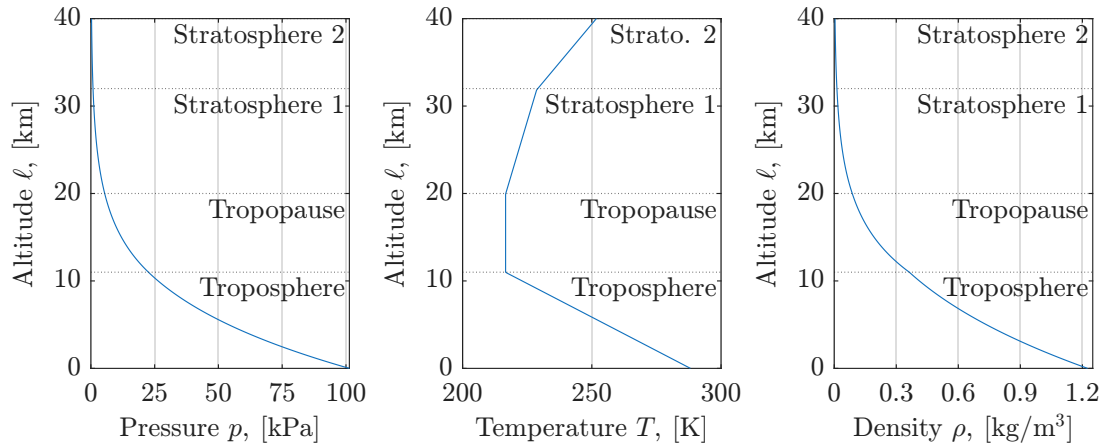


Figure 3.3: Air variations with altitude of the model atmosphere(r_x).

The following atmosphere model is used to predict the barometric pressure measurements and the aerodynamic effects accurately. The air at altitude r_x is modeled according to the US standard atmosphere [NASA76, Stengel04] as

atmosphere(r_x) :

$$\ell = \frac{R_0 r_x}{R_0 - r_x}, \quad (3.20)$$

$$T = T_b - k_b(\ell - \ell_b), \quad (3.21)$$

$$p = \begin{cases} p_b \left(1 - \frac{k_b}{T_b}(\ell - \ell_b)\right)^{\frac{g_0}{Rk_b}}, & \text{for } k_b \neq 0 \\ p_b \exp\left(\frac{-g_0}{RT_b}(\ell - \ell_b)\right), & \text{for } k_b = 0 \end{cases} \quad (3.22)$$

$$\rho = \frac{p}{RT}, \quad (3.23)$$

$$c = \sqrt{\gamma RT}, \quad (3.24)$$

where the index b indicates a base value from Tab. 3.1. The geographic altitude above sea level r_x is converted to the geopotential altitude ℓ for the atmospheric calculations, using the mean Earth radius $R_0 = 6356.77$ km. The gravitational acceleration is $g_0 = 9.81 \frac{\text{m}}{\text{s}^2}$. The adiabatic index of air is $\gamma = 1.4$, and the specific gas constant of air is $R = 287.0579 \frac{\text{J}}{\text{kgK}}$.

Table 3.1: US Standard Atmosphere base data [NASA76]

Layer	Altitude $\ell_b[\text{m}]$	Pressure $p_b[\text{Pa}]$	Temperature $T_b[\text{K}]$	Lapse rate $k_b [\text{K/m}]$
Troposphere	0	101 325	288.15	0.0065
Tropopause	11 000	22 632.1	216.65	0
Stratosphere 1	20 000	5474.9	216.65	-0.0010
Stratosphere 2	32 000	868.02	228.65	-0.0028

Air data and incidence angles

The airspeed of the rocket is $V_A = \|\mathbf{v}\|$, which assumes no wind velocity as any wind will act as disturbance on the system. The dynamic pressure $\bar{p}$ and the Mach number Ma are defined as

$$\bar{p} = \frac{1}{2}\rho V_A^2, \quad \text{Ma} = \frac{V_A}{c}. \quad (3.25)$$

The dynamic pressure is the predominant factor in the aerodynamic forces, while the Mach number influences the aerodynamic coefficients. The incidence of the rocket relative to the oncoming airstream is characterized with the angle-of-attack α and the angle-of-sideslip β as shown in Fig. 3.4. Both α and β are defined by the velocity of the rocket in the body-fixed frame as

$$\alpha = \text{atan2}(v_z, v_x), \quad \beta = \text{atan2}(v_y, v_x). \quad (3.26)$$

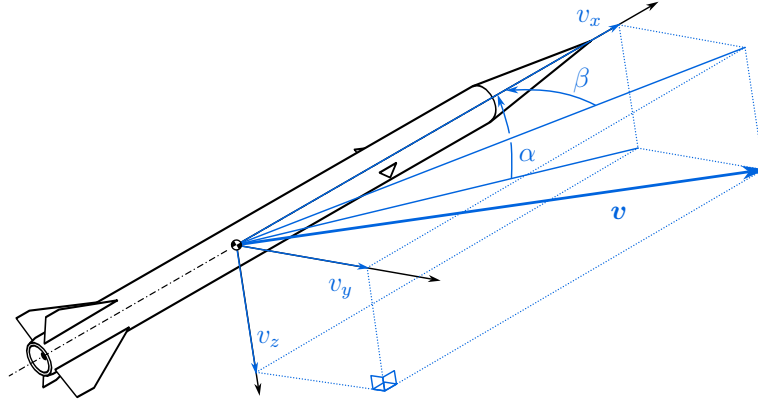


Figure 3.4: Diagram of the incidence angles α and β in the body-fixed frame.

3.4 Aerodynamic forces and torques

External forces and torques act on the rocket in flight, perturbing it from free motion. Aside from thrust most of these forces and torques stem from aerodynamics. As described in Sect. 3.2 only the torque formulations are used for the controller and estimator models. The drag force and thrust will therefore not be modeled. External torques are mainly produced by lateral forces acting on the rocket body away from the center of gravity as

$$\mathcal{T}^B = \mathbf{d}^B \times \mathcal{F}^B, \quad (3.27)$$

with the center of pressure location $\mathbf{d}^B$. The aerodynamic forces are defined as

$$\mathcal{F}_i = \bar{p} A C_i(\text{shape}, \alpha, \delta, \dots) \quad (3.28)$$

with the dynamic pressure $\bar{p}$ and the reference area A . The coefficients C_i depend on the body geometry and the direction of flow. Individual centers of pressure and force coefficients of body components, such as the nosecone and tail fins, can be predicted from geometry [Barrowman67].

Normal forces and torques

Any non-zero incidence produces a normal force acting on the overall center of pressure $\mathbf{d}_{\text{cp}}^B = [d_{\text{cp}} \ 0 \ 0]^\top$. With the approximation $C_N(\alpha) \approx C_{N_\alpha} \alpha$ made by Barrowman, the normal forces and the resulting pitch and yaw torques are

$$\mathcal{F}_N = \bar{p} A_{\text{ref}} C_{N_\alpha} \begin{bmatrix} 0 \\ -\sin(\beta) \\ -\sin(\alpha) \end{bmatrix} \quad \mathcal{T}_N = \bar{p} A_{\text{ref}} d_{\text{cp}} C_{N_\alpha} \begin{bmatrix} 0 \\ \sin(\alpha) \\ -\sin(\beta) \end{bmatrix} \quad (3.29)$$

The values of the coefficients, areas, and moment arms for the combined rocket body are listed in Tab. 2.1. The normal force coefficient derivative C_{N_α} and the axial center of pressure location d_{cp} were computed with the Barrowman method using OPENROCKET [NiskanenGoolPfeiffer25]. A detailed derivation of these parameters exceeds the scope of this thesis.

Canard forces and torques

Canard actuation by an angle δ redirects the oncoming air as shown in Fig. 3.5, producing a normal force $\mathcal{F}_{\text{can},n}$ and an axial force $\mathcal{F}_{\text{can},d}$. For the small deflections $|\delta| \leq 10^\circ$ of the canard surfaces the normal force is proportional to the actuation angle [Stengel04], so the coefficient of the canard normal force is

$$C_L(\delta) = C_{L_\delta} \delta. \quad (3.30)$$

Thin airfoil theory for the coefficient C_{L_δ} is treated in the appendix in Sect. C.1.

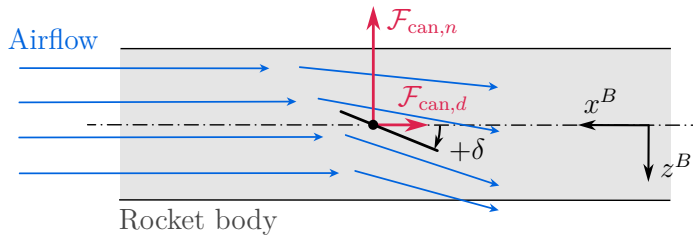


Figure 3.5: Canard actuation. Only the forces of a single canard are shown.

As the canards are on the opposite sides of the rocket body and actuate by equal and opposite angles, the normal forces $\mathcal{F}_{\text{can},n}$ cancel out and a net moment about the roll axis remains. With the radial center of pressure d_{can} of one canard the torque vector of the canards is thus

$$\sum \mathcal{F}_{\text{can},n} = 0 \quad \implies \quad \mathcal{T}_{\text{can}} = \bar{p} A_{\text{can}} d_{\text{can}} C_{L_\delta} \begin{bmatrix} \delta & 0 & 0 \end{bmatrix}^\top. \quad (3.31)$$

The values for area and moment arm are listed in Tab. 2.1. This formulation does not account for interaction of the canard downwash with the tail fins. The effect of vortices causing roll reversal are difficult to predict with available tools [LandersHallAuman00] so they are not modeled. Additionally, the freely rolling tail fins of the demonstrator should mitigate any adverse torques [Blair78, LesieutreQuijano14]. The axial forces $\mathcal{F}_{\text{can},d}$ are not modeled in detail, but it is to note that actuation therefore increases the drag force on the rocket.

3.5 Actuator model

The drive train is made up of a servo motor and gearing. The servo motor acts as a position control element, which provides the required torque to reach and hold the commanded shaft angle. The internal control loop of the servo has at least second order dynamics, angle and rate limits, and dead-time.

A reasonable approximation of the actuator dynamics for controller design would be a first or second order linear system. For simplification the former is chosen, which has been shown to be a sufficient approximation [FresconiEtAl14]. The first order system with the canard angle δ and the command δ_{cmd} has the form

$$\dot{\delta} = -\alpha_{\delta}(\delta - \delta_{\text{cmd}}). \quad (3.32)$$

The response is fitted to the measured actuator response by adjusting the time constant α . The response was measured with the canard encoder, without load on the shaft. For model fitting the different initial and target angles of each run were normalized. The model shown in Fig. 3.6 is fitted with a time constant of $\alpha_{\delta} = \frac{1}{0.03\text{s}}$ and is discretized. While the measured responses are quite fast, they exhibit a long dead time of about the same length as the settling time, which also varies strongly between test trials. When accounting for the sampling rate, the worst-case dead time delay is about $T_{\text{delay}} = 0.05\text{s}$. The control loop must account for the phase loss caused by the dead time.

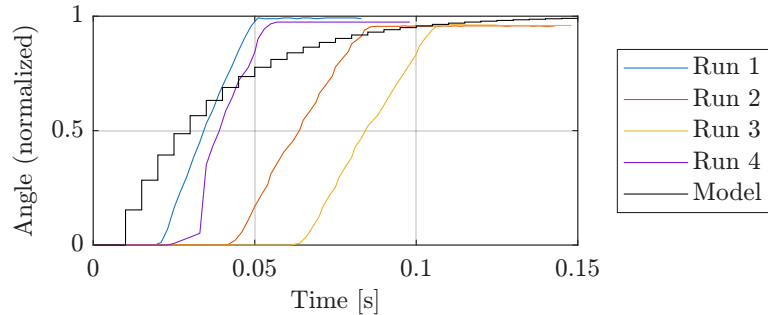


Figure 3.6: Measured normalized actuator responses and first order model.

Chapter 4

Controller

In this chapter the controller is designed to stabilize the rocket's roll motion. The control scheme is introduced, the roll control laws are described, and responses of the linear loop are analyzed.

A linear quadratic regulator (LQR) is chosen as the feedback controller. This optimal control method is used as minimal actuation is desired to lower aerodynamic drag, and because the canard actuation angle is limited. Furthermore, LQR controllers exhibit good robustness, and LQR roll control was already successfully demonstrated [FresconiEtAl14]. As the focus is on stabilization, the control architecture is a two-degree-of-freedom design as illustrated in Fig. 4.1.

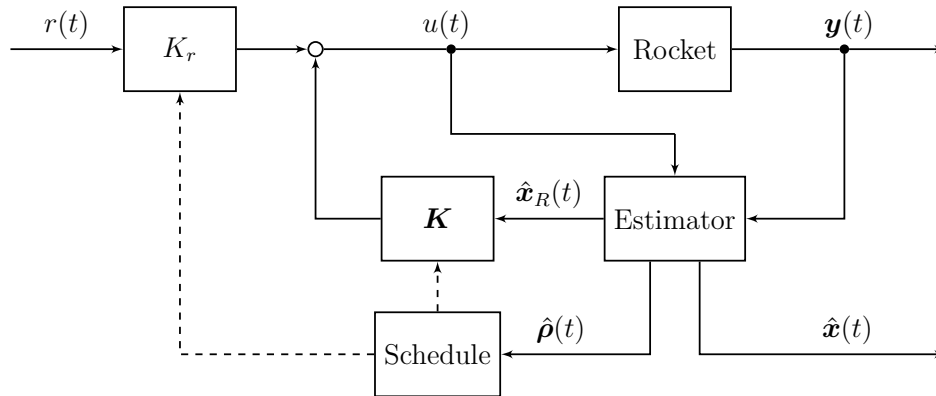


Figure 4.1: Block diagram of the control loop, a 2-DOF estimator-based state feedback with gain scheduling.

The LQR is a state feedback controller and thus requires the roll state $\mathbf{x}_R$ for feedback with gains $\mathbf{K}$. An estimator is used to determine the rocket states from measurements $\mathbf{y}$. Independent design of the feedback controller and estimator is possible under the separation principle, even if the system is partly nonlinear [AtassiKhalil98]. The estimator is therefore treated in Chap. 5.

To account for the varying nature of the rocket flight dynamics, the controller selects the feedforward and feedback gains from a gain schedule according to the flight condition $\boldsymbol{\rho}$, which is provided by the estimator. The feedback regulates the roll state to zero, but for evaluation of the control performance some step responses are desired. A reference roll program with multiple steps is implemented as the signal r , which is scaled by a feedforward gain K_r for tracking.

4.1 Optimal control

The problem of optimal control is extensively treated in literature [BrysonHo75, Chachuat06, Tewari11, Werner21b], therefore only a short overview of linear quadratic optimal control is provided here.

Given are the controllable linear state-space model on the interval $t \in [0, \infty]$

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad (4.1)$$

and the quadratic cost functional

$$V(\mathbf{x}(0), \mathbf{u}(\cdot)) = \int_0^\infty (\mathbf{x}^\top(t)\mathbf{Q}\mathbf{x}(t) + \mathbf{u}^\top(t)\mathbf{R}\mathbf{u}(t)) dt \quad (4.2)$$

with weighting matrices $\mathbf{Q} \succeq \mathbf{0}$ and $\mathbf{R} \succ 0$ which penalize the state error and the control effort, respectively. The optimization problem

$$\begin{aligned} \min_{\mathbf{u}(\cdot)} V(\mathbf{x}(t), \mathbf{u}(\cdot)) \\ \text{s.t. Eq. (4.1)} \end{aligned} \quad (4.3)$$

has the solution with optimal value V^* and control $\mathbf{u}^*$ [Chachuat06]

$$V^*(t) = \mathbf{x}^\top(t)\mathbf{P}(t)\mathbf{x}(t), \quad (4.4)$$

$$\mathbf{u}^*(t) = -\mathbf{R}^{-1}\mathbf{B}^\top\mathbf{P}(t)\mathbf{x}(t). \quad (4.5)$$

When the time horizon is infinite, $\mathbf{P}$ is constant and satisfies the algebraic Riccati equation [Chachuat06, Werner21b]

$$\mathbf{P}\mathbf{A} + \mathbf{A}^\top\mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top\mathbf{P} + \mathbf{Q} = \mathbf{0}, \quad (4.6)$$

the linear quadratic regulator has thus the optimal control law

$$\mathbf{u}(t) = \mathbf{K}\mathbf{x}(t) \quad \text{with} \quad \mathbf{K} = -\mathbf{R}^{-1}\mathbf{B}^\top\mathbf{P}. \quad (4.7)$$

The closed-loop dynamics are then

$$\dot{\mathbf{x}}(t) = (\mathbf{A} + \mathbf{B}\mathbf{K})\mathbf{x}(t). \quad (4.8)$$

The solution of $\mathbf{P}$ to Eq. (4.6) and the feedback gain $\mathbf{K}$ can be computed with MATLAB's `lqr()` function. The closed loop is guaranteed to be stable and robust with margins of at least -6 dB to $+\infty$ dB in gain and 60° in phase [SafonovAthans77]. However this guarantee is only valid for linear systems with full state measurements, margins can be lost in combination with estimation [Doyle78]. However the control law is still useful for controlling nonlinear and observed systems, as the robustness properties can be at least partially preserved by adjusting estimator gains [DoyleStein79], this will be treated in Chap. 5.

4.2 Reduced controller model

As the canards can only produce a pure roll torque on the vehicle the whole attitude dynamics are under-actuated. The complex dynamics derived in Chap. 3 are therefore projected onto the controllable subspace, the resulting model used for controller design is then a simplified linear parameter-varying (LPV) system with a reduced state $\mathbf{x}_R$ and a parameter vector $\boldsymbol{\rho}$.

The controllable subspace consists of the roll dynamics. The altitude and velocity vector are not controllable, but as a flight condition the dynamic pressure is an element of $\boldsymbol{\rho}$. Rockets are rotational symmetric and typically have the diagonal inertia $\mathbf{J} = \text{diag}[J_{xx} \ J_{yy} \ J_{yy}]$ as explained for Eq. (3.13). Due to the equal moments of inertia on the lateral axes the roll acceleration is written by

$$\dot{\omega}_x = \frac{\mathcal{T}_x}{J_{xx}} = L_\delta \delta \quad \text{with} \quad L_\delta(\bar{p}, C_{L_\delta}) = \bar{p} C_{L_\delta} \frac{A_{\text{can}} d_{\text{can}}}{J_{xx}}. \quad (4.9)$$

A derivation of the relationship between roll angle and roll rate is provided in Sect. C.3. The roll angle and its approximate time derivative are

$$\phi = \text{atan2}\left(2(q_w q_x + q_y q_z), (q_w^2 - q_x^2 - q_y^2 + q_z^2)\right), \quad \dot{\phi} = \omega_x. \quad (4.10)$$

The controllable dynamics are composed of three state variables for the parameter-varying roll model, thus the roll state and parameter vector are

$$\mathbf{x}_R = [\phi \ \omega_x \ \delta]^\top, \quad \boldsymbol{\rho} = [\bar{p} \ C_{L_\delta}]. \quad (4.11)$$

The commanded canard angle δ_{cmd} is the control input, the resulting LPV state-space model with first order actuator dynamics is

$$\dot{\mathbf{x}}_R = \mathbf{A}(\boldsymbol{\rho}) \mathbf{x}_R + \mathbf{B} \delta_{\text{cmd}}, \quad (4.12)$$

$$\text{with} \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & L_\delta(\boldsymbol{\rho}) \\ 0 & 0 & -\alpha_\delta \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \alpha_\delta \end{bmatrix}. \quad (4.13)$$

Controllability is a sufficient condition for the LQR method. Therefore the roll model is checked for controllability, with the controllability matrix

$$\mathbf{C} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \end{bmatrix}. \quad (4.14)$$

This matrix has full rank for $\alpha_\delta > 0$ and for all $L_\delta \neq 0$, i.e. the system is controllable for every flight condition except for zero dynamic pressure ($\|\mathbf{v}\| = 0$ or $\rho = 0$), or a zero coefficient of lift ($C_{L_\delta} = 0$), which confirms intuition as aerodynamic control surfaces need airflow to be effective.

4.3 Scheduled control law

The desired canard angle $u = \delta_{\text{cmd}}$ is computed by the linear control law

$$u(t) = \mathbf{K}(\hat{\boldsymbol{\rho}}) \hat{\mathbf{x}}_R(t) + K_r(\hat{\boldsymbol{\rho}}) r(t), \quad (4.15)$$

where $\hat{\mathbf{x}}_R$ and $\hat{\boldsymbol{\rho}}$ are the reduced state and flight condition as determined by the state estimator. The gains $\mathbf{K} = \begin{bmatrix} K_\phi & K_{\omega_x} & K_\delta \end{bmatrix}$ are determined by performing the LQR method on the linear model as

$$\mathbf{K}(\boldsymbol{\rho}) = -\text{lqrd}(\mathbf{A}(\boldsymbol{\rho}), \mathbf{B}, \mathbf{Q}_c, R_c, dt). \quad (4.16)$$

The MATLAB function `lqrd()` is used for synthesizing a feedback gain for discrete-time execution on a processor with sampling time dt , using a zero-order-hold exact discretization of the continuous-time model. The regulator is tuned by adjusting the weighting $\mathbf{Q}_c$ and R_c . As shown in Sect. 4.1, the matrix $\mathbf{Q}_c$ determines the cost of any state error, while R_c weighs the expended control effort. To follow a roll program, the roll angle reference signal $r(t)$ is scaled by the feedforward gain K_r . This gain is calculated by the static inversion of the linear model

$$K_r(\boldsymbol{\rho}) = \frac{-1}{\mathbf{C} [\mathbf{A}(\boldsymbol{\rho}) + \mathbf{BK}(\boldsymbol{\rho})]^{-1} \mathbf{B}}, \quad (4.17)$$

such that the steady-state gain from reference to output is 1. The tracking variable is chosen by $\mathbf{C}$, which produces a linear combination of the reduced state variables. With $\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$, the reference value is tracked by the roll angle. Alternatively, roll rate tracking would be done by $\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$.

The weights are chosen heuristically as

$$\mathbf{Q}_c = \text{diag} \left(\begin{bmatrix} 10 & 4 & 2 \end{bmatrix} \right), \quad R_c = 300 + 0.006 \bar{p}, \quad (4.18)$$

where the control input penalty is scaled with the dynamic pressure. This scaling is done to reduce the loop sensitivity for high control authority, and to further reduce actuation-induced drag at high speeds.

Gain scheduling

As the model of the roll dynamics is dependent on the flight condition parameters, a gain-scheduling scheme is used. This is common practice in aerospace applications for dealing with varying flight conditions [Buschek99, Theis23]. Design points $\boldsymbol{\rho}_{i,j} \in \{\boldsymbol{\rho}\}$ of the scheduling parameters are selected on a grid [TheisEtAl15] with grid points

$$\{\boldsymbol{\rho}\} = \{\bar{p}\} \times \{C_{L_\delta}\}, \quad (4.19)$$

$$\begin{aligned} \{\bar{p}\} : \quad & \bar{p} \in [250 \text{ Pa} \quad 600 \text{ kPa}] , \quad n_{\bar{p}} = 200 , \\ \{C_{L_\delta}\} : \quad & C_{L_\delta} \in [-15 \quad 15] \setminus 0 , \quad n_{C_L} = 30 . \end{aligned} \quad (4.20)$$

The condition $L_\delta = 0$ is omitted from the tuning process, as the system is not controllable then. A large number of design points $n_{\bar{p}}, n_{C_L}$ are chosen to ensure sufficient margins at all flight conditions. The control gains for all design points are computed beforehand and stored in a look-up table, the resulting surfaces are shown in Fig. 4.2. During flight, gains are linearly interpolated from the lookup

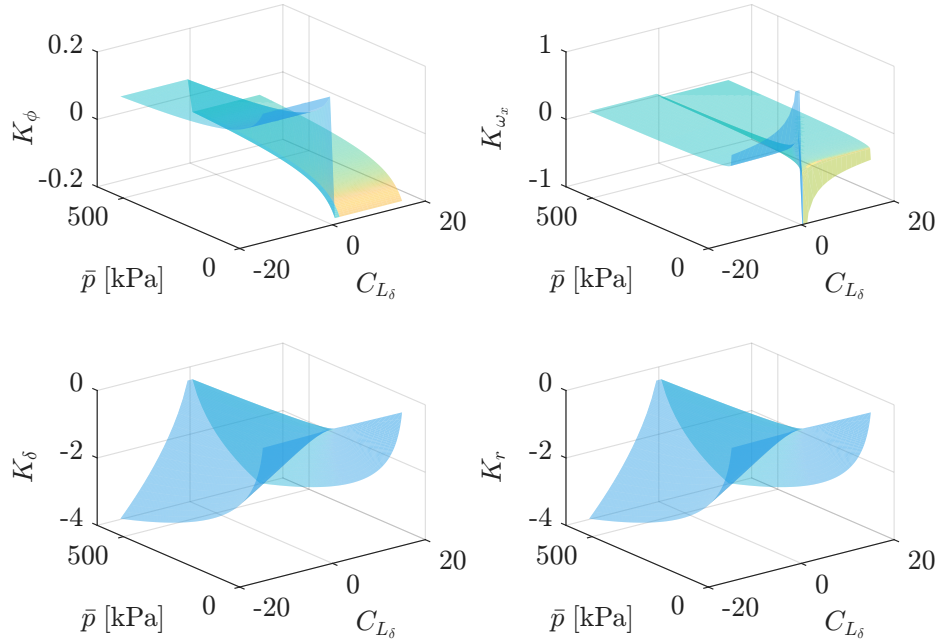


Figure 4.2: Scheduling surfaces along dynamic pressure $\bar{p}$ and canard coefficient of lift C_L . Feedback gains K_ϕ, K_p, K_δ and feedforward gain K_r .

table using the parameters $\hat{\boldsymbol{\rho}}$ projected from the current state estimate. As the LQR method produces guaranteed robustness margins, the interpolation between two close design points should always produce a stable loop. The estimate has

next lower and upper neighbors (points 1 and 2) on the look-up table grid. With $K_{ij} = K(\bar{p}_i, C_{L,j})$ the bilinear interpolation for each element of K is

$$K(\bar{p}, C_L) = \frac{1}{(\bar{p}_2 - \bar{p}_1)(C_{L,2} - C_{L,1})} \begin{bmatrix} \bar{p}_2 - \bar{p} \\ \bar{p} - \bar{p}_1 \end{bmatrix}^\top \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} C_{L,2} - C_L \\ C_L - C_{L,1} \end{bmatrix}. \quad (4.21)$$

4.4 Response analysis

The closed-loop step response is shown in Fig. 4.3, for the flight conditions $r_x = 1000$ m, $C_{L_\delta} = 4$, and velocity variations between 30 m/s (green) and 900 m/s (red). The response time is sufficient for settling any step reference or disturbance within 3 s for most dynamic pressures. At low velocities of 30 m/s the control authority is strongly reduced and the rise time prolongs to about 5 s. For the lowest control authority the canard angle just reaches the mechanical $\pm 10^\circ$ limit of the canard system (Sect. 2.3).

The Bode plots of the open-loop system are shown in Fig. 4.4 for the same flight conditions as before. With the chosen controller gains the highest crossover frequency of the open-loop, and resulting closed-loop bandwidth, is just below of $\omega_B = 6$ rad/s. In practice when using exact discretization the sampling frequency of a discrete controller should be at least 10 times larger than the closed-loop bandwidth [Werner21a]. Thus the sampling frequency should be about $f_s = 10 \omega_B / \pi \approx 20$ Hz. The estimator dynamics should be faster than the closed-loop system dynamics to mitigate the influence of estimator dynamics in the loop [Werner21a]. Furthermore, to mitigate aliasing effects a faster sampling frequency is chosen such that the open-loop gain has fallen below -40 dB. For the chosen weights this occurs at a frequency of about $\omega_{-40 \text{ dB}} \approx 300 \dots 600$ rad/s. The sampling frequency should thus be $f_s = \omega_{-40 \text{ dB}} / \pi \approx 100 \dots 200$ Hz.

The controller requires sufficient phase margins to compensate for the dead time of the loop, which is predominantly due to the actuator delay. The actuator response from Sect. 3.5 showed a dead time of $T_{\text{delay}} = 0.05$ s, reducing the phase by $\varphi_{\text{loss}} = -\omega T_{\text{delay}}$. At the largest airspeed of 900 m/s the delayed phase has the lowest response, this is shown in orange in Fig. 4.4. Accounting for the phase loss incurred by the actuator delay, the phase margin is about $\varphi_{\text{margin}} = 54^\circ$. By design the gain margin of the linear system is about 40 dB. However, when accounting for the phase loss the gain margin shrinks to about 14.5 dB, as the phase crosses -180° already at $\omega_{-180^\circ} = 24$ rad/s. Both gain and phase margin indicate a robust control loop, but show sensitivity to actuator delay at high dynamic pressures. The estimator and active scheduler are not included in this loop, and their nonlinear adaptive behavior makes further robustness analysis quite difficult.

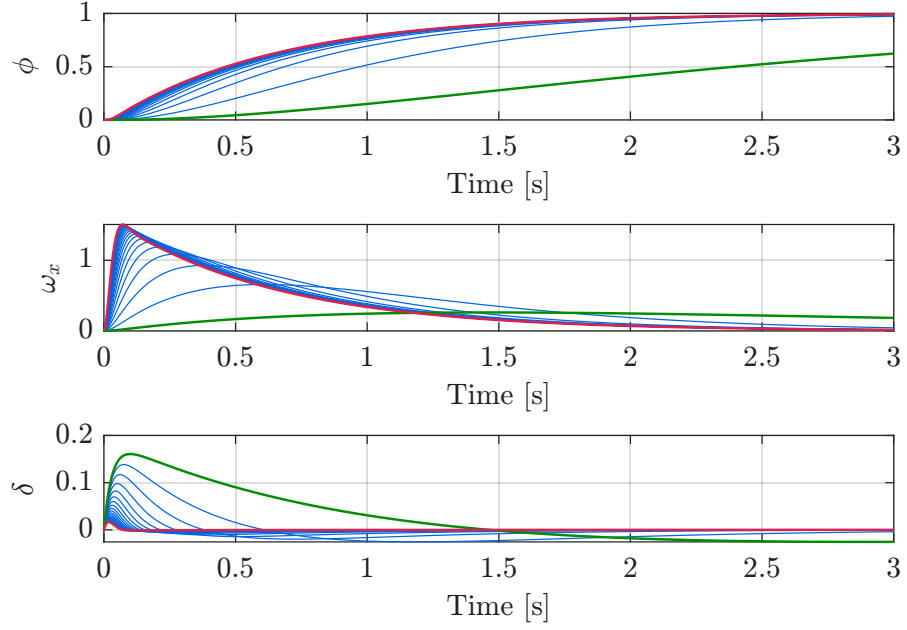


Figure 4.3: Step responses of the closed-loop system. Velocity variations between 30 m/s (green) and 900 m/s (red). Base unit in rad.

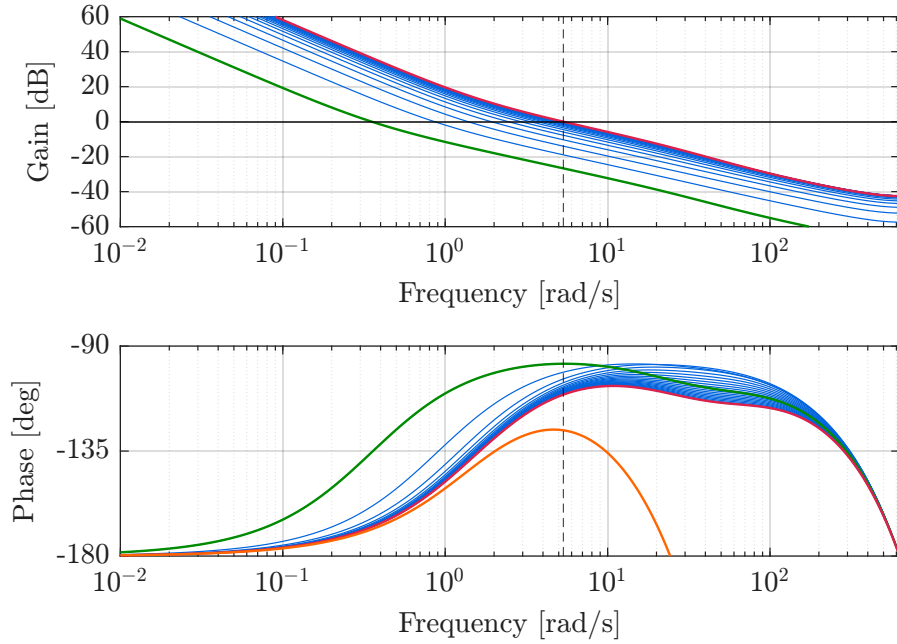


Figure 4.4: Frequency response of the open-loop system, with a sampling frequency of $f_s = 200$ Hz. Highest crossover frequency marked with dashed line. Velocity variations between 30 m/s (green) and 900 m/s (red). Worst delayed response in orange.

Chapter 5

Estimator

In this chapter the estimator is designed to determine the rocket state from control inputs and onboard sensor measurements. The estimator consists of two stages, the flight filter treated in Sect. 5.3, and the pad filter treated in Sect. 5.4. The pad filter is executed while the rocket is stationary on the launch pad, and determines the initial state and sensor biases.

An extended Kalman filter (EKF) is chosen for the flight filter, as this optimal estimation method minimizes the uncertainty of the estimate while accounting for nonlinear dynamics. Furthermore, as an observer the Kalman filter is able to determine states which cannot be measured directly. The EKF is chosen over other nonlinear Kalman filter variations for simplicity [LewisXiePopa08]. The EKF is a Bayes filter, so each estimate consists of a mean state $\hat{\mathbf{x}}$ and an associated belief $\mathbf{P}$.

The estimator is implemented on a microprocessor, so the filter is computed in discrete-time. As explained in Sect. 4.4 the estimator dynamics should be faster than the controller dynamics, and the sampling frequency must be selected accordingly. Based on the frequency response the sampling rate is chosen as $dt = 0.005$ s to about $dt = 0.01$ s.

5.1 Optimal estimation

Optimal estimation is a dual problem to optimal control, and is also extensively treated in literature [Kalman60, SmithSchmidtMcGee62, BrysonHo75, LewisXiePopa08]. The detectable discrete state-space model of the plant is given by

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{G}\mathbf{u}_k + \mathbf{w}_k, \quad \mathbf{y}_k = \mathbf{H}\mathbf{x}_k + \mathbf{n}_k, \quad (5.1)$$

where $\mathbf{w}_k$ is the process noise and $\mathbf{n}_k$ is the measurement noise. The noise vectors are assumed to be white, zero-mean Gaussian with covariances $\mathbf{Q} \succeq 0$ and $\mathbf{R} \succ 0$

such that they satisfy [Werner21a]

$$\begin{aligned} \mathbf{w} &\sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \implies \mathbb{E}[\mathbf{w}_k \mathbf{w}_{k+1}^\top] = \mathbf{Q}, \\ \mathbf{n} &\sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \implies \mathbb{E}[\mathbf{n}_k \mathbf{n}_{k+1}^\top] = \mathbf{R}. \end{aligned} \quad (5.2)$$

The estimator is structured as a state observer, i.e. a copy of the plant model with measurement feedback. This dynamical system is described in state-space by [LewisXiePopa08, StevensJohnsonLewis15]

$$\hat{\mathbf{x}}_{k+1} = \mathbf{F}\hat{\mathbf{x}}_k + \mathbf{G}\mathbf{u}_k + \mathbf{K}(\mathbf{y}_k - \mathbf{H}\hat{\mathbf{x}}_k), \quad (5.3)$$

with the mean estimate $\hat{\mathbf{x}}$. The variable to be regulated is the estimation error $\boldsymbol{\varepsilon} = \mathbf{x} - \hat{\mathbf{x}}$ with the dynamics

$$\boldsymbol{\varepsilon}_{k+1} = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1} = (\mathbf{F} - \mathbf{K}\mathbf{H})\boldsymbol{\varepsilon}_k + \mathbf{w}_k - \mathbf{K}\mathbf{n}_k, \quad (5.4)$$

which are similar to Eq. (4.8). The error covariance of an estimate is

$$\mathbf{P}_k = \mathbb{E}[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top] = (\mathbf{F} - \mathbf{K}\mathbf{H})\mathbf{P}_{k-1}(\mathbf{F} - \mathbf{K}\mathbf{H})^\top + \mathbf{Q} + \mathbf{K}\mathbf{R}\mathbf{K}^\top. \quad (5.5)$$

As derived by Kalman the minimization problem is [Kalman60]

$$\begin{aligned} \min_{\mathbf{K}} \text{trace}(\mathbf{P}) \\ \text{s.t. Eq. (5.4)}. \end{aligned} \quad (5.6)$$

At the optimal solution and for an infinite time horizon, $\mathbf{P}$ satisfies the filter algebraic Riccati equation in steady state $\mathbf{P}_{k+1} = \mathbf{P}_k$ as [Kalman60]

$$\mathbf{P} = \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q} - \mathbf{F}\mathbf{P}\mathbf{H}^\top[\mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}]^{-1}\mathbf{H}^\top\mathbf{P}\mathbf{F}^\top. \quad (5.7)$$

This is the discrete version of the control algebraic Riccati equation (Eq. (4.6)), and similarly the solution is the optimal feedback gain

$$\mathbf{K} = \mathbf{P}\mathbf{H}^\top\mathbf{R}^{-1}, \quad (5.8)$$

which implicitly depends on the current gain. To solve the gain explicitly, the covariance propagation of a discrete Kalman filter can be written as a two-step process by reformulating the gain, as [LewisXiePopa08]

$$1. \quad \mathbf{P}_{k+1}^- = \mathbf{F}\mathbf{P}_k\mathbf{F}^\top + \mathbf{Q}, \quad (5.9)$$

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}^- \mathbf{H}^\top [\mathbf{H}\mathbf{P}_{k+1}^- \mathbf{H}^\top + \mathbf{R}]^{-1}, \quad (5.10)$$

$$2. \quad \mathbf{P}_{k+1}^+ = (\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H})\mathbf{P}_{k+1}^-. \quad (5.11)$$

These derivations are for the linear discrete Kalman filter. The extended Kalman filter was developed to observe a system of nonlinear equations of motion, by linearizing the motion and perturbation equations around the current mean estimate [SmithSchmidtMcGee62]. For the EKF the constant matrices $\mathbf{F}, \mathbf{H}$ are replaced by first order linearizations of the nonlinear models in the covariance process. The EKF is generally only quasi-optimal, as convergence behavior is not global for most nonlinear systems, and because the assumption of Gaussian noise propagation requires linear systems [LewisXiePopa08].

5.2 Estimator models

The rocket model developed in Chap. 3 can be written as a system of differential equations, represented in state-space as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \quad \mathbf{y} = \mathbf{h}(\mathbf{x}) \quad (5.12)$$

with the state $\mathbf{x}$, input $\mathbf{u}$, and output $\mathbf{y}$. The dynamics model requires the rocket attitude $\mathbf{q}$, the angular velocity in the body-fixed frame $\boldsymbol{\omega}$, the velocity in the body-fixed frame $\mathbf{v}$, and altitude r_x . Position is reduced to altitude, as longitude and latitude are not observable without GPS. The entire state vector is then

$$\mathbf{x} = [\mathbf{q}^\top \quad \boldsymbol{\omega}^\top \quad \mathbf{v}^\top \quad r_x \quad C_{L_\delta} \quad \delta]^\top, \quad (5.13)$$

by including the canard coefficient of lift C_{L_δ} and the canard angle δ . As a state the coefficient of lift can be corrected by measurement feedback, which is relevant if roll reversal occurs in flight or in case the airfoil model is inaccurate. As described in Sect. 3.2, the measured acceleration $\underline{\mathbf{a}}$ is used as an input to the model instead of the measurement feedback. The input vector is

$$\mathbf{u} = [\delta_{\text{cmd}} \quad \underline{\mathbf{a}}^\top]^\top, \quad (5.14)$$

with the controller command to the actuator δ_{cmd} . The actual canard angle is represented as the state δ to account for actuator dynamics. Excluding the accelerometer, the sensor data includes the measured angular velocity $\underline{\boldsymbol{\omega}}$ from the gyroscope, the magnetic field $\underline{\mathbf{m}}$ from the magnetometer, the static air pressure $\underline{p}$ from the barometer, and the encoder signal $\underline{\delta}$. These are combined in the model output vector

$$\mathbf{y} = [\underline{\boldsymbol{\omega}}^\top \quad \underline{\mathbf{m}}^\top \quad \underline{p} \quad \underline{\delta}]^\top. \quad (5.15)$$

The model of the rocket dynamics predicts the time derivative of the state vector $\dot{\hat{\mathbf{x}}}$, using the current state estimate and input. To render the model detectable in case the canards are inactive, the canard coefficient state includes first order dynamics. This converges the state estimate to the theoretical value (appendix Sect. C.1). The relevant equations from the model are: attitude Eq. (3.8), angular velocity Eq. (3.13), velocity Eq. (3.12), altitude Eq. (3.11), and actuator Eq. (3.32).

With the accelerometer input $\mathbf{a} = (\underline{\mathbf{a}} - \mathbf{b}_a) - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{d}_A)$ from Eq. (3.15), the dynamics are

$$\dot{\hat{\mathbf{x}}} = \mathbf{f}(\hat{\mathbf{x}}, \mathbf{u}) = \begin{bmatrix} \frac{1}{2} \mathbf{q} * \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \\ \mathbf{J}^{-1}(\boldsymbol{\mathcal{T}} - \boldsymbol{\omega} \times \mathbf{J} \boldsymbol{\omega}) \\ \mathbf{a} - \boldsymbol{\omega} \times \mathbf{v} + \mathbf{S} \mathbf{g}^G \\ (\mathbf{S}^\top \mathbf{v})_x \\ -\alpha_c(C_L - C_{L, \text{theory}}) \\ -\alpha_\delta(\delta - \delta_{\text{cmd}}) \end{bmatrix}. \quad (5.16)$$

The measurement models predict the sensor outputs $\hat{\mathbf{y}}$ from the current state $\hat{\mathbf{x}}$. The relevant equations are: angular velocity Eq. (3.16), magnetic field Eq. (3.17), atmospheric pressure Eq. (3.18) and Eq. (3.22), and the encoder measurement Eq. (3.19). The combined measurement models are

$$\hat{\mathbf{y}} = \mathbf{H}(\hat{\mathbf{x}}) = \begin{bmatrix} \boldsymbol{\omega} + \mathbf{b}_\omega \\ \mathbf{S} \mathbf{m}_E^G + \mathbf{b}_m \\ \text{atmosphere}(r_x)_p + b_p \\ \delta + b_\delta \end{bmatrix}. \quad (5.17)$$

Integration and Linearization

For a discrete-time update the continuous-time dynamics have to be solved by integration. Here explicit Euler is chosen for simplicity and low computational effort. The discrete update function $\mathbf{f}_D$ is

$$\begin{aligned} \mathbf{x}_k &= \int_0^t \dot{\mathbf{x}}(\mathbf{x}, \mathbf{u}) d\tau \\ &\approx \mathbf{f}_D(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}) = \mathbf{x}_{k-1} + dt \cdot \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}). \end{aligned} \quad (5.18)$$

The EKF requires first-order linearizations of the models for the covariance process. This requires that the filter is computed at sufficiently short time intervals such that the state does not change greatly between each computation step. These linearizations are the sensitivities of the functions $\mathbf{f}_D$ and $\mathbf{h}$ to variations of the state, i.e. the Jacobian matrices

$$\mathbf{F}_D(\mathbf{x}, \mathbf{u}) = \frac{\partial \mathbf{f}_D}{\partial \mathbf{x}} = \mathbf{I} + dt \mathbf{F}(\mathbf{x}, \mathbf{u}) \quad \text{with} \quad \mathbf{F}(\mathbf{x}, \mathbf{u}) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}, \quad (5.19)$$

$$\mathbf{H}(\mathbf{x}) = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}. \quad (5.20)$$

The partial derivatives were derived analytically and are listed in App. B.

Observability

To test if the filter converges at a given state, the linearized system is examined for observability. The observability matrix is

$$\mathcal{O} = \begin{bmatrix} \mathbf{H} \\ \mathbf{H}\mathbf{F}_D \\ \mathbf{H}\mathbf{F}_D^2 \\ \vdots \\ \mathbf{H}\mathbf{F}_D^{n_x} \end{bmatrix}. \quad (5.21)$$

Full observability requires $\text{rank}(\mathcal{O}) = n_x$, with the number of states $n_x = 13$. Based on the structure of the Jacobians (appendix Eq. (B.2) and Eq. (B.4)), full rank is obtained if the rocket has a nonzero velocity and the canards are actuating. For inactive canards the rank drops to 12, as the canard coefficient is not observable without roll acceleration. For zero velocity the rank drops to 10, as without aerodynamics or GPS only the velocity component in the altitude direction is observable. The system is still detectable when the observability falls in rank, as the velocity and canard coefficient states are stable in the model.

5.3 EKF algorithm

The extended Kalman filter propagates the mean state estimate $\hat{\mathbf{x}}$ in discrete time steps, along with the error covariance matrix $\mathbf{P}$. A detailed description of the EKF is given in [LewisXiePopa08]. Due to the partial nonlinearity of the model functions, both $\mathbf{P}$ and $\mathbf{K}$ must be calculated in real time as new measurements become available. One iteration of the discrete EKF algorithm is:

Predict:

$$\hat{\mathbf{x}}_k^- = \mathbf{f}_D(\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}), \quad \text{normalize } \hat{\mathbf{q}}_k^- \quad (5.22)$$

$$\mathbf{P}_k^- = \mathbf{F}_D(\hat{\mathbf{x}}_k^-, \mathbf{u}_{k-1})\mathbf{P}_{k-1}^+\mathbf{F}_D^\top(\hat{\mathbf{x}}_k^-, \mathbf{u}_{k-1}) + \mathbf{Q}_e \quad (5.23)$$

Correct:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^\top(\hat{\mathbf{x}}_k^-) \left[\mathbf{H}(\hat{\mathbf{x}}_k^-) \mathbf{P}_k^- \mathbf{H}^\top(\hat{\mathbf{x}}_k^-) + \mathbf{R}_e \right]^{-1} \quad (5.24)$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k \left[\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{x}}_k^-) \right], \quad \text{normalize } \hat{\mathbf{q}}_k^+ \quad (5.25)$$

$$\mathbf{P}_k^+ = \left[\mathbf{I} - \mathbf{K}_k \mathbf{H}(\hat{\mathbf{x}}_k^-) \right] \mathbf{P}_k^- \left[\mathbf{I} - \mathbf{K}_k \mathbf{H}(\hat{\mathbf{x}}_k^-) \right]^\top + \mathbf{K}_k \mathbf{R}_e \mathbf{K}_k^\top. \quad (5.26)$$

The prediction steps consist of the nonlinear discrete-time dynamics Eq. (5.22) and the associated covariance update in Eq. (5.23) using the local sensitivity

of the dynamics $\mathbf{F}(\hat{\mathbf{x}}_k^-)$. To ensure valid attitude quaternion operations, the quaternion is normalized after prediction and correction. The filter gain $\mathbf{K}_k$ is computed in Eq. (5.24) using the covariance $\mathbf{P}_k^-$ and the local sensitivity of the measurement model $\mathbf{H}(\hat{\mathbf{x}}_k^-)$. Correction feedback in Eq. (5.25) is performed with the innovation, i.e. the difference between actual measurements $\mathbf{y}_k$ and modeled measurements $\hat{\mathbf{y}}_k$. The covariance correction in Eq. (5.26) uses the Joseph stabilized form to ensure $\mathbf{P}$ remains positive semidefinite, even if the gain $\mathbf{K}_k$ is suboptimal [LewisXiePopa08].

As multiple redundant sensors are available, multiple correction steps are performed sequentially. If a measurement is missed due to sensor fault or firmware timing, the prediction step can be repeated as well. Correcting sequentially is also useful to reduce the size of the matrix inversion in Eq. (5.24). All sensors have their own measurement model with different bias values.

The EKF is initialized by setting the state and covariance estimates to

$$\hat{\mathbf{x}}_0^+ = \mathbf{x}_0, \quad \mathbf{P}_0 = \mathbf{0}, \quad (5.27)$$

where $\mathbf{x}_0$ is computed while the rocket is stationary on the rail by the pad filter from Sect. 5.4. The covariance of the initial state is set to zero as the state on the launch pad is very certain. The quaternion norm must be satisfied for initialization, an exactly vertical launch is $\mathbf{q}_0 = [1 \ 0 \ 0 \ 0]^\top$.

Filter tuning

As shown in Eq. (5.2), the tuning matrices $\mathbf{Q}_e \succeq 0$ and $\mathbf{R}_e \succeq 0$ represent the expected noise variances of the dynamics and measurements processes. The EKF is tuned by setting the entries of these matrices, which could be filled to account for cross-covariance between the states or measurements (e.g. a higher altitude or higher velocity lowers the accuracy of the pressure measurement). However, this is often difficult to characterize, and in practice the tuning matrices are often filled diagonally [Werner21a]. Typically, the values in $\mathbf{R}_e$ represent the noise behavior of the sensors, where the entries are usually taken directly from sensor datasheets. The matrix $\mathbf{Q}_e$ can be adjusted according to the expected “noise” in the state prediction process, which in practice represents expected disturbances and thus trust in the accuracy of the state prediction.

Closed-loop robustness depends on the margins of the combined rocket-estimator-controller loop. Although a separated estimator-controller design can have sufficient robustness margins [SafonovAthans77, AtassiKhalil98], Doyle showed that using a state estimator inside the control loop could lead to vanishing margins [Doyle78]. To recover some of the robustness margins of the optimal controller (Sect. 4.1), a simple loop transfer recovery method is employed [DoyleStein79].

With the sensitivity of the control input $\mathbf{G}_{\text{cmd}}$, the adjusted weighting is

$$\mathbf{Q}_e \rightarrow \mathbf{Q}_e + \sigma_{\text{cmd}}^2 \mathbf{G}_{\text{cmd}} \mathbf{G}_{\text{cmd}}^\top \quad \text{with} \quad \mathbf{G}_{\text{cmd}} = \frac{\partial \mathbf{f}}{\partial \delta_{\text{cmd}}}, \quad (5.28)$$

which is tuned by choosing some fictitious noise $\sigma_{\text{cmd}}^2 \geq 0$ acting on the control input channel. When this noise becomes large, the optimal and robust state feedback properties are asymptotically recovered [DoyleStein79, Werner21b].

After preliminary simulation trials the weighting matrices $\mathbf{Q}$ and $\mathbf{R}$ are chosen as

$$\mathbf{Q}_e = \text{diag} \left[[\mathbf{1}_{1,4}] \text{E}^{-10}, \quad [1\text{E}^{-3}, 0.1, 0.1], \quad [\mathbf{1}_{1,3}] \text{E}^{-6}, \quad 1\text{E}^{-3}, \quad 0.3, \quad 0.1 \right] \quad (5.29)$$

$$\mathbf{R}_e = \text{diag} \left[[\mathbf{1}_{1,3}] \text{E}^{-6}, \quad [\mathbf{1}_{1,3}] \text{E}^{-2}, \quad 1, \quad 1\text{E}^{-3} \right] \quad (5.30)$$

5.4 Pad filter

This filter shall determine accurate values for the initial state $\mathbf{x}_0$ and the sensor biases $\mathbf{b}$. The bias determination reduces dependency of the filtered attitude on the noisy magnetometer by improving the accuracy of gyroscope integration. As the rocket acceleration off the pad may exceed accelerometer limits, good barometer bias determination increases accuracy in the velocity estimates during the boost phase. Shortly before liftoff, the pad filter hands over the latest initial state and sensor biases to the EKF.

While the rocket is stationary on the launch pad, the pad filter runs an exponential low-pass filter

$$\mathbf{s}_{k+1} = \alpha \mathbf{y}_k + (1 - \alpha) \mathbf{s}_k, \quad (5.31)$$

with the current sensor data $\mathbf{y}_k$ and filtered data $\mathbf{s}_k$ (initialized as $\mathbf{s}_0 = \mathbf{y}_0$). Filtering should take place for several minutes to attenuate any rocket movement caused by fill operations or wind, while the low-pass smooths the noisy sensor measurements. The gain α is chosen according to the expected time frame of initialization. For the launch of Aurora, the time constant was chosen as $\alpha = 0.0002$. The filtered data $\mathbf{s}$ is used for the remaining computations of the pad filter, whenever sensor data is required.

As the rocket is stationary, the following assumptions can be made.

1. The body angular velocity will be zero, and thus the attitude is constant.
2. The velocity of the rocket will be zero, and thus the altitude is constant.
3. The acceleration of the rocket relative to the geographic frame will be zero, thus the acceleration relative to the inertial frame is gravity.

The initial state thus consists of $\mathbf{v}_0 = \mathbf{0}$, $\boldsymbol{\omega}_0 = \mathbf{0}$, and the canards are straightened to $\delta_0 = 0$. The initial attitude quaternion $\mathbf{q}_0$ is determined from the filtered accelerometer data

$$\mathbf{a} = -\mathbf{S}(\mathbf{q}_0) \mathbf{g}^G. \quad (5.32)$$

The initial orientation the rocket is not exactly vertical, as the launch rail is intentionally inclined. The initial orientation of the body-fixed frame relative to the geographic frame is defined with $q_x = 0$, such that the inclination of the rocket can be uniquely determined (with $\|\mathbf{a}\| = 1$) by

$$q_{w,0} = \sqrt{\frac{1 + a_x}{2}}, \quad q_{x,0} = 0, \quad q_{y,0} = \frac{a_z}{2q_{w,0}}, \quad q_{z,0} = \frac{-a_y}{2q_{w,0}}. \quad (5.33)$$

A derivation of this inclinometer method is given in the appendix in Sect. C.2. The launch altitude $r_{x,0}$ can be set to a constant value as the launch location and its altitude above sea level is known. Furthermore, $C_{L,0}$ can be set to a constant value determined from the airfoil model.

Earth's magnetic field vector is determined from $\mathbf{q}_0$ as

$$\mathbf{m}_E^G = \mathbf{S}^\top(\mathbf{q}_0) \underline{\mathbf{m}}, \quad (5.34)$$

such that the estimation does not depend on magnetic field models. This requires however a prior calibration of the magnetometer hard iron bias $\mathbf{b}_m$. The accelerometer measurement is used for inclination determination, thus the accelerometer biases $\mathbf{b}_a$ cannot be determined on the launch pad and instead must be calibrated beforehand. The gyroscope bias is the filtered angular velocity as the rocket is stationary, the same applies to the encoder bias. Because the launch altitude $r_{x,0}$ is known, any barometer bias can be calculated by comparison with the atmosphere model. From the filtered measurements the biases are therefore

$$\mathbf{b}_\omega = \underline{\boldsymbol{\omega}}, \quad b_p = \underline{p} - \text{atmosphere}(r_{x,0})_p, \quad b_\delta = \underline{\delta}. \quad (5.35)$$

Chapter 6

Simulation

Estimator and controller performance is evaluated in simulation. The simulation framework is described in Sect. 6.1, and results of individual simulations are discussed in Sect. 6.2. Next to single simulations, used to test specific behavior, Monte-Carlo simulations were conducted. Shown in Sect. 6.3, these varied simulations are used to verify the control loop robustness. To validate the embedded code the onboard processor is tested with hardware-in-the-loop simulations, which are discussed in Sect. 6.4.

6.1 ClosedRocket simulation framework

Available sounding rocket simulation software, such as OPENROCKET [NiskanenGoolPfeiffer25] and RASAERO [RogersCooper19], are widely used for the development of amateur rockets and provide adequate aeroprediction capabilities for statically stable configurations. However, they lack support for movable aerodynamic surfaces or realistic sensor and actuator models, limiting their use for closed-loop controller development. To overcome these limitations and enable testing of a canard-based roll controller, a new high-fidelity six-degree-of-freedom simulation framework was developed in MATLAB/SIMULINK. Aptly titled CLOSEDROCKET [BreuerDolinaYanKlassen25], this simulation supports closed-loop control by integrating the controller/estimator processor module as shown in Fig. 6.1. Unlike previous tools, it is capable of simulating actuators, sensors, and aerodynamics with moving surfaces. CLOSEDROCKET also supports Monte-Carlo simulations with random variations in wind, rocket parameters, sensor disturbances, and actuator performance.

The simulation model is implemented in SIMULINK, enabling the use of its numerical solvers for the continuous-time flight dynamics. For modularity and maintainability, the simulation is structured as a hierarchical architecture. Each major part of the simulation, such as the forces, environmental model, or sensor em-

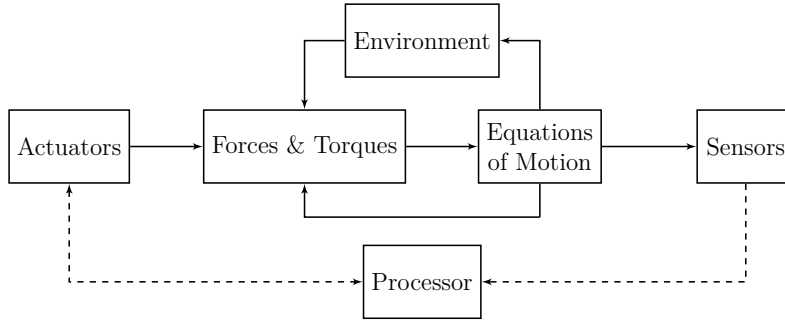


Figure 6.1: Block diagram of general signal flow of the simulation. Solid lines are continuous-time, dashed lines are discrete-time.

ulation, has its own module, which is then further structured into layers with component submodules. This structure isolates individual physical effects (e.g. forces from fins or canards) and thus simplifies debugging, and allows isolated testing and upgrades in model fidelity.

To provide a proper validation environment for the controller and estimator, the simulation models were in part developed with different approaches than those of the model in Chap. 3. Standard components, such as the 6DOF equations of motion, are sourced from the MATLAB Aerospace Toolbox to reduce implementation effort. Aerodynamic forces and torques are computed using Barrowman’s method [Barrowman67], which is validated experimentally for sub- and supersonic flow. Canard aerodynamics are similar to fins in the Barrowman method, but the algorithms were specifically modified to support moving surfaces. Thrust forces and the varying mass distribution are taken from static engine tests and propulsion simulations. Interactions of the rocket with the launch rail are modeled with acceleration constraints, and recovery events after apogee are simulated with parachute forces. Wind disturbances include constant wind, random turbulence, and discrete gusts. The environment is modeled with the International Standard Atmosphere, the geodetic reference WGS-84 with its gravity model, and the International Geomagnetic Reference Field. Sensors include accelerometers, rate gyroscopes, magnetometers, and barometers. They are subject to randomly varied bias instability and stochastic noise, saturation limits, and non-aligned mounting. Actuator models have second-order dynamics with limits, dead-time delay, backlash, and noisy encoder measurements.

Verification of the simulation models was done by comparing the flight trajectories to results from OPENROCKET for identical flight configurations. Further details on the simulation models and validation exceed the scope of this work.

6.2 Simulation results

The following simulations are conducted. Firstly, a nominal flight is simulated with low wind and other disturbances. This serves as a baseline for the flight simulation and as a controller and estimator performance test under imperfect conditions. Secondly, the behavior of a controlled flight is compared with an uncontrolled flight where coning motion is present due to disturbances. Lastly, a simulation with realistic conditions with regard to the expected flight at Launch Canada is conducted, with a focus on the controller performance. This work focuses on roll control for coning mitigation during ascent, so the focus is on the time from launch to apogee.

6.2.1 Aurora nominal

For a nominal flight, wind disturbances are low, roll reversal does not happen, and all sensors are reliable. The simulation of the flight is shown in Fig. 6.2. Liftoff occurs at $t = 10$ s and the boost phase lasts for 10 s, unpowered ascent continues until apogee at $t = 60$ s. Shortly after apogee the drogue parachute is deployed,

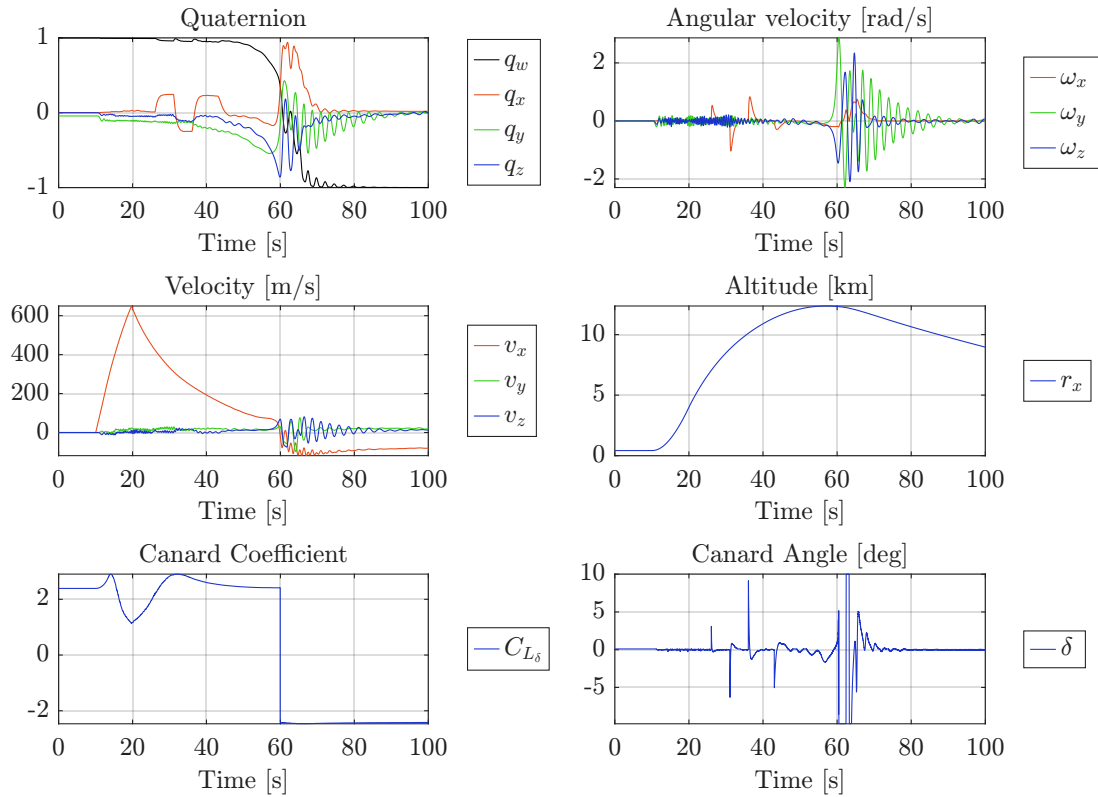


Figure 6.2: Nominal simulated states of the closed-loop system.

resulting in chaotic tumbling of the rocket for about 10 s before the attitude begins to settle. Finally, the rocket descends slowly under parachute. As this takes several minutes and the main mission of the controller is the ascent, only the first 20 s of descent are shown. The controller is active for the entire flight, regulating disturbances acting on the roll motion. The roll behavior shows essentially no oscillations or overshoot during ascent, as indicated by the quaternion parameter q_x . For better evaluation of the controller performance a reference roll program is tracked, starting at $t = 27$ s and lasting until $t = 45$ s. Multiple roll angle steps are performed to 0.5 rad and -0.5 rad, which the controller is able to track well.

The performance of the estimator is evaluated by comparing the “ground truth” states $\mathbf{x}$ and estimated states $\hat{\mathbf{x}}$ in simulation. To quantify the estimation accuracy the estimation error $\boldsymbol{\varepsilon} = \mathbf{x} - \hat{\mathbf{x}}$ is used, shown in Fig. 6.3. The largest relative errors are in the canard coefficient $C_{L\delta}$, this is likely due to different implementations of the aerodynamics between estimator model and flight simulation. This state becomes unobservable when $\delta = 0$ or $\|\mathbf{v}\| = 0$, which can be seen shortly after liftoff or during parachute deployment. The velocity $\mathbf{v}$ shows the largest error during parachute deployment. There are no direct measurements available for the velocity, so this estimate relies largely on accelerometer integration. Furthermore, as the rocket is essentially in free fall during most of the flight the accelerometer

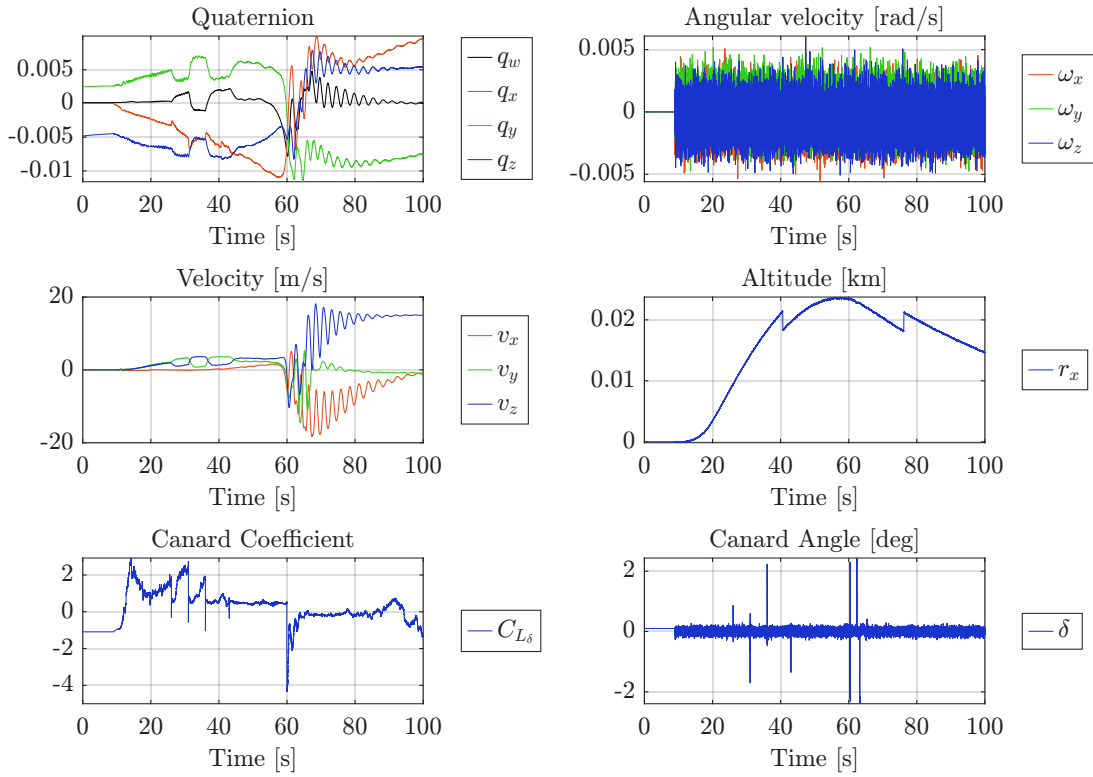


Figure 6.3: Nominal estimation error of the simulated closed-loop system.

cannot measure the gravity vector, and any gravitational acceleration is modeled using the attitude $\mathbf{q}$. Both the body rates and the altitude are estimated very well, which is due to the rate gyroscope measuring the former directly, and the atmosphere model directly coupling barometer measurement and altitude. Due to the estimator using a different atmosphere model than the simulation, a small altitude error of about 20 m is present at apogee. The attitude quaternion is also estimated quite well due to gyroscope rate integration, where the bias is determined on the launch pad, and any remaining error is slowly corrected by the magnetometer. Any initial attitude error is due to using the accelerometer as an inclinometer, which is subject to small acceleration biases. The canard angle estimate matches well with the true value, because approximated dynamics are used and direct correction from the encoder is available.

6.2.2 Coning analysis

The difference between controlled and uncontrolled flight is investigated. In this simulation, liftoff occurs at time $t = 0$ s, the boost phase lasts until $t = 10$ s, and apogee is reached at about $t = 47$ s. A roll disturbance is introduced by a small canted surface, effectively a canard a deflected by 0.5° . Additionally, the center of pressure of the tail fins is laterally offset from the center of gravity of the rocket by 10 mm. For clarity, only the relevant ground truth states of the simulation are shown in the figures. In the uncontrolled flight shown in Fig. 6.4, these disturbances cause a spin up of the roll rate ω_x during the boost phase

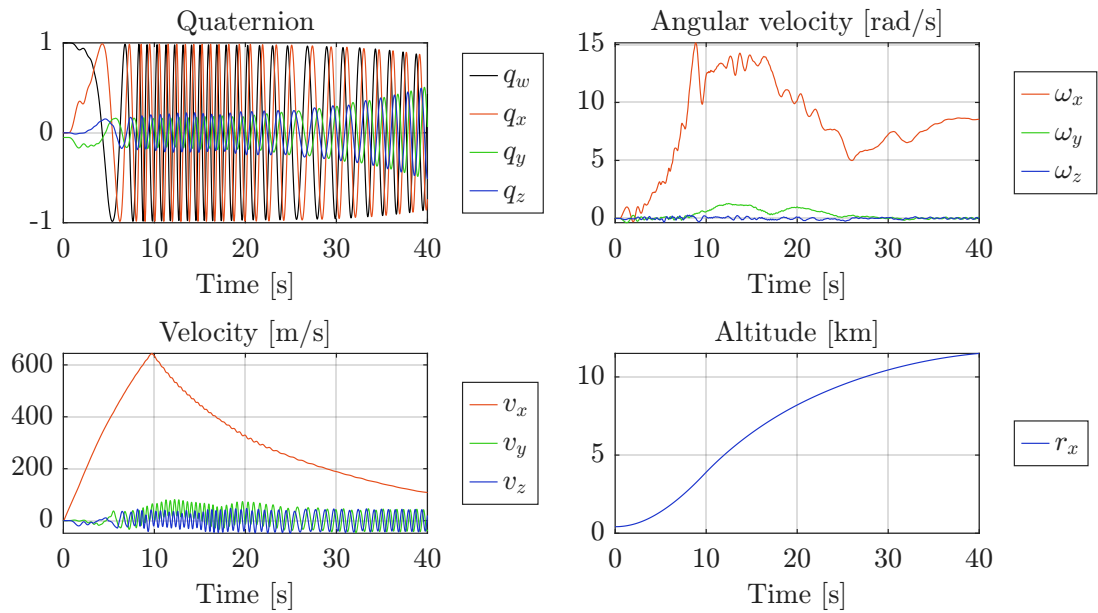


Figure 6.4: Simulation of uncontrolled flight, with large disturbances.

and a stable coning motion. The rocket enters the the lock-in region at about $t = 8$ s, but escapes at roughly $t = 16$ s. During unpowered ascent the coning motion increases, a precession is indicated by the oscillations of the quaternion parameters q_y, q_z and the lateral velocities v_y, v_z . Even though the ascent velocity slows approaching apogee, the coning motion does not subside.

The controlled flight with otherwise identical simulation parameters is shown in Fig. 6.5. In this case the roll rate does not increase during the boost phase, and stays below ± 1 rad/s for the entire flight. There is only little oscillation in the quaternion and lateral velocity for the entire ascent, and the lateral velocities are overall significantly lower. The roll program is successfully performed during unpowered ascent, as indicated by the the quaternion parameter q_x and the roll rate responses ω_x . Due to the disturbances in the rocket geometry the controller is very active, as indicated by fluctuations in the canard angle and roll rate responses.

Comparing the uncontrolled with the controlled flight, the controller is clearly able to regulate the roll motion of the rocket. Furthermore, active roll control inhibits the oscillatory coning motion and mitigates the risk of roll lock-in.

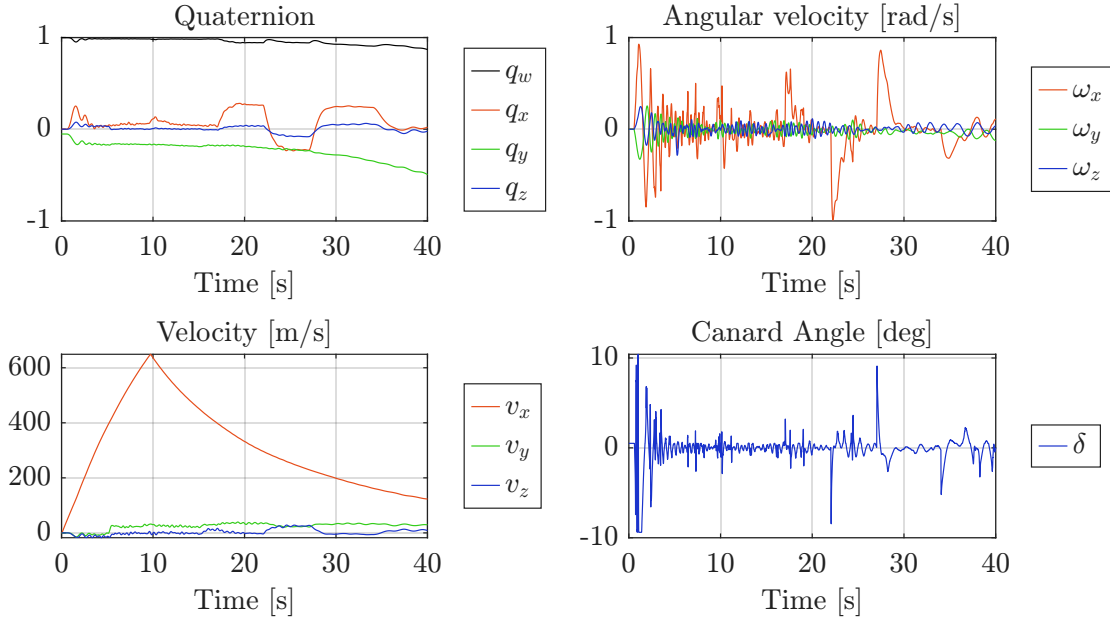


Figure 6.5: Simulation of controlled flight, with large disturbances and roll program.

6.2.3 Expected flight attitude

As Aurora is to be launched at Launch Canada, all systems have to comply with the rules of the competition. According to the design, test & evaluation guide, active flight control systems shall only be actuating after the boost phase of the engine has ended [LC25]. This requirement is implemented as a timer which starts at liftoff and counts down to the expected burnout time. The simulation of the rocket flight is shown in Fig. 6.6.

Liftoff occurs at $t = 10$ s. The boost phase lasts for 10 s, therefore engine burnout and timer-based controller activation occurs at $t = 20$ s. During the boost phase, a roll spin up to $\omega_x \approx 8$ rad/s occurs due to slight asymmetries of the rocket geometry. As soon as the controller starts actuating the canards, the roll rate is reduced drastically within a few seconds. The multi-step roll program starts at $t = 27$ s and lasts until $t = 45$ s. The controller is able to track the roll angle steps as indicated by the quaternion parameter q_x , albeit with a few oscillations.

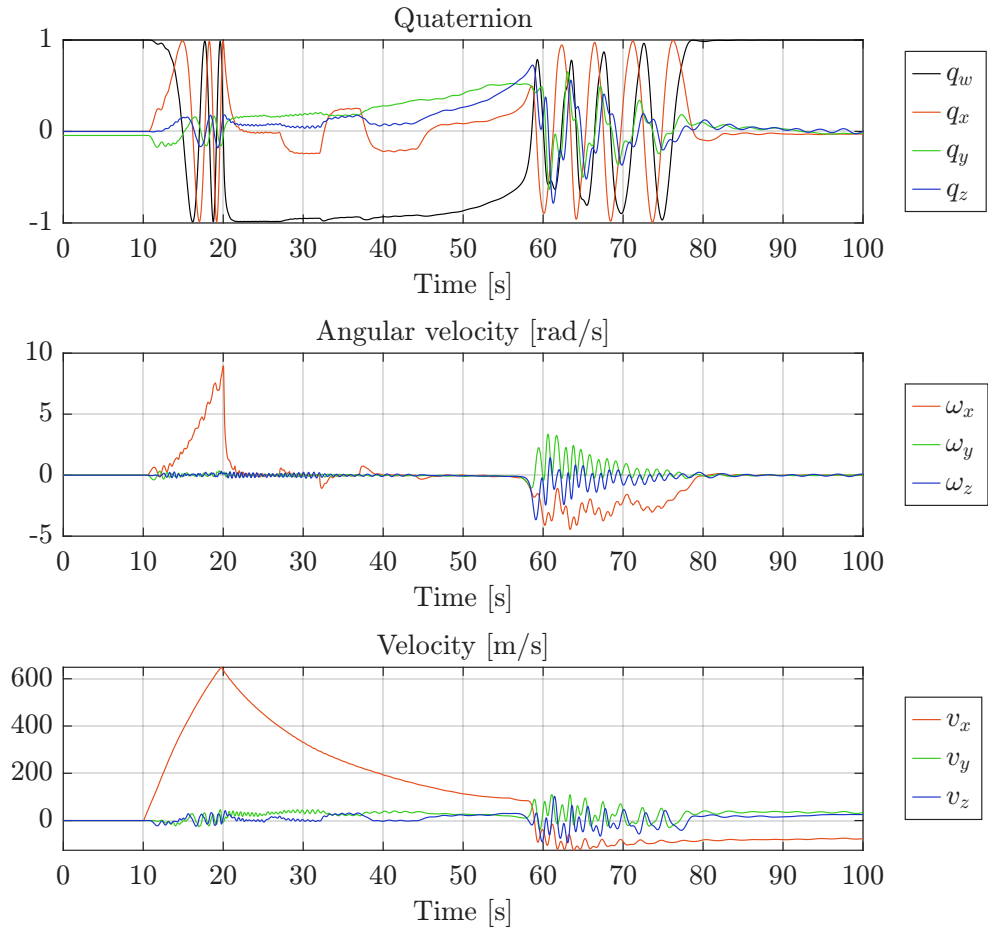


Figure 6.6: Simulation of the attitude of the expected Aurora flight.

At $t = 58$ s apogee is reached and the drogue parachute is deployed. This causes a few seconds of tumbling as the rocket is pulled around by the parachute, after which the rocket settles into a steady descent and oscillations subside. Due to the low airspeed and erratic tumbling the controller is not able to keep up with roll regulation during the deployment phase. However, after the tumbling has settled roll control is successfully continued.

The estimation errors for the attitude $\mathbf{q}$, angular velocity $\boldsymbol{\omega}$, and velocity $\mathbf{v}$ are shown in Fig. 6.7. Both quaternion and angular velocity exhibit very small errors, while the velocity error becomes largest during the parachute deployment phase, where accelerometer integration is challenging due to fast tumbling.

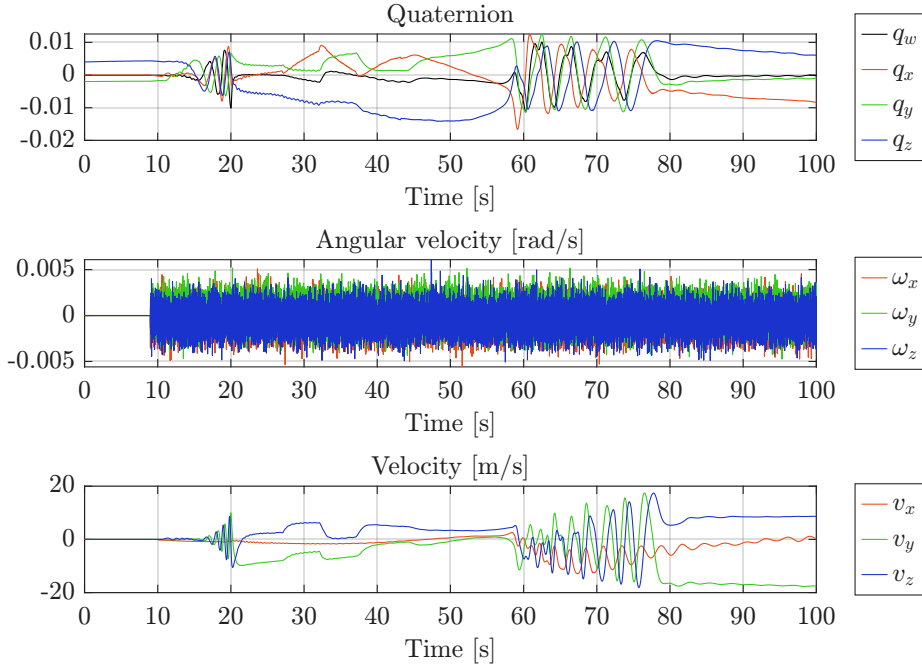


Figure 6.7: Attitude estimation error of the expected Aurora flight.

6.3 Monte-Carlo simulations

Monte Carlo simulations are conducted to assess the robustness of the estimator and controller, and their sensitivity to uncertain initial conditions, imperfect sensor measurements, and plant-model mismatch. Random variations of wind speed, roll reversal effects, and more can be applied for this purpose. Variations are selected on normal distributions by choosing a mean, lower bound, and upper bound for each parameter. The random sampling uses a standard deviation of $\sigma = \frac{\text{Upper}-\text{Lower}}{4}$ around the mean, such that 95 % of variations lie between the bounds, and 5 % of variations randomly select either boundary value.

The following simulations were run in a batch of 200, with the parameter variations listed in Tab. 6.1. Next to wind disturbances and varying engine performance, canard parameters and roll disturbances are varied. Fin cant is the slight tilt of the tail fins, such that they produce a roll torque under forward velocity. Most notable is the canard coefficient factor, which is used to quantify uncertainty in the aerodynamics. A factor of -1 is an effective roll reversal, which simulates canard downwash interacting with the fins resulting in an opposite roll torque.

Table 6.1: Parameter variations for Monte-Carlo simulations.

Parameter	Mean	Lower	Upper	Unit
Engine thrust factor	1	0.85	1.05	.
Wind constant speed	10	0	20	m/s
Wind gust amplitude	10	0	30	m/s
Canard coefficient factor	1	-1	2	.
Canard backlash	0.25	0	1	deg
Fin cant angle	0.1	0	0.5	deg

In simulation, liftoff occurs at $t = 10$ s, the controller is activated at $t = 15$ s, and the boost phase lasts until $t = 20$ s. Apogee is reached between $t = 50$ s and $t = 60$ s, depending on the engine thrust variations.

The angular velocity responses of the rocket are shown in Fig. 6.8. The large differences in attitude dynamics caused by varying fin and canard parameters are distinct. Some spin up occurs during the boost phase which subsides after the controller is activated. It is apparent that roll control keeps the roll rate small for at least 80 % of cases, but some simulations show a significantly increased

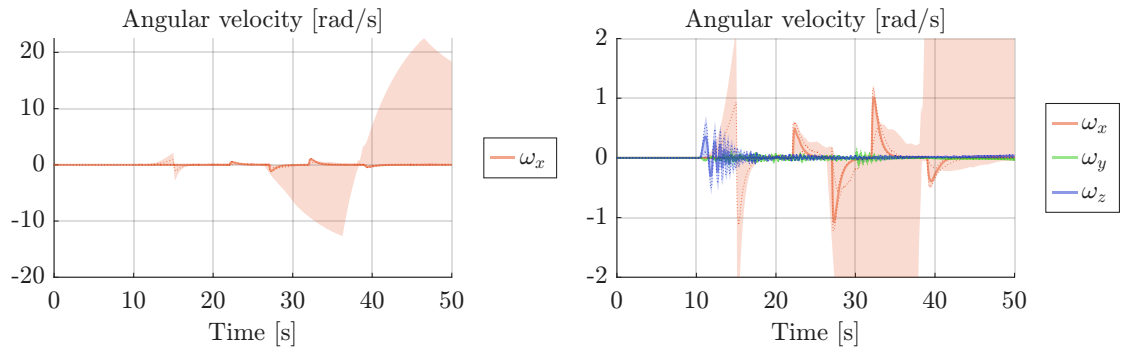


Figure 6.8: Angular velocity in Monte-Carlo simulations. Just the roll rate on the left, magnification on the right. Solid line is the median value, dotted lines mark the 80th percentile, transparent fill includes 95% of all simulations.

roll rate. Large roll is present when the random variations result in an effective canard coefficient near zero and a large fin cant. Additionally, the rocket loses control authority near apogee where the dynamic pressure is low. During the unpowered ascent, the rocket roll angle follows the roll program smoothly for at least 80 % of simulations.

The estimation errors of these simulations are shown in Fig. 6.9. For 80 % of simulations the estimation errors are low, similar to the nominal flight profile, and all error responses are bounded. As expected the errors of the attitude quaternion, angular velocity, altitude, and canard angle are small and partly show just sensor noise. For no more than 20 % of simulations, larger errors are present for v_y, v_z as indicated by thin spikes in the velocity plot, starting at $t = 40$ s. Notably, the canard coefficient does get estimated successfully even when roll reversal effects are present, which effectively inverts the coefficient value.

The covariance matrix $\mathbf{P}$ represents the current uncertainty of the EKF process in the state estimate. It is used to evaluate the stability of the estimator, its norm

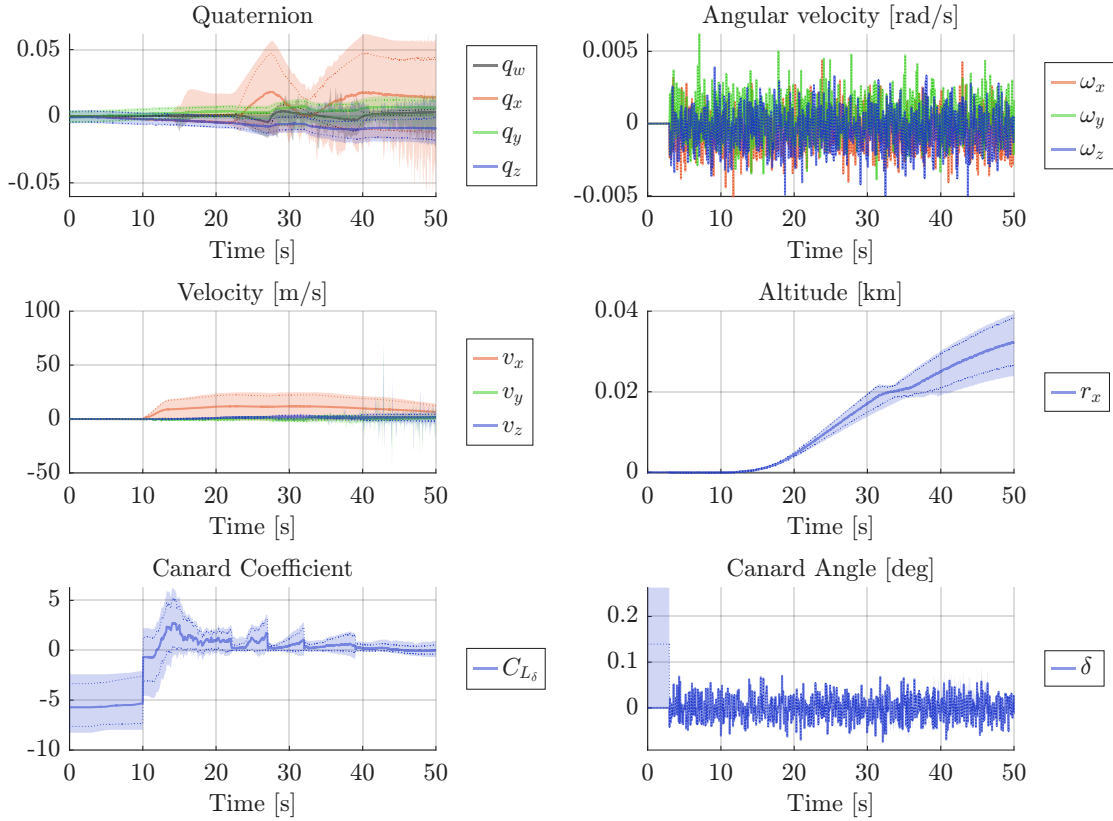


Figure 6.9: Estimation error of the Monte-Carlo simulations. Solid line is the median value, dotted lines mark the 80th percentile, transparent fill includes 95% of all simulations.

is shown in Fig. 6.10. The operation $\text{max norm}(\mathbf{P})$ returns the maximum of the largest singular value, absolute row sum, and absolute column sum. Essentially this is largest uncertainty, or sum of uncertainties, in the current iteration of the EKF. Notably, the covariance norm rises when the canards are approximately straight as shown for $t \geq 5$ s and $t \geq 40$ s. When the rocket accelerates or the canard actuates the covariance norm reduces, as shown by the boost phase at $t = 15$ s, and at the roll program steps at $t = 27$ s, $t = 33$ s and $t = 38$ s. This indicates that most uncertainty lies in the coefficient estimate, which has a similar pattern in its estimation error. In Sect. 5.2 it was shown that the system becomes partially unobservable when either the canard is inactive or the rocket has zero velocity. Simulations match the theory, as covariance increases are most notable when the state is near the unobservable region. However, the maximum norm of $\mathbf{P}$ stays bounded for 95% of simulations, indicating otherwise robust estimation.

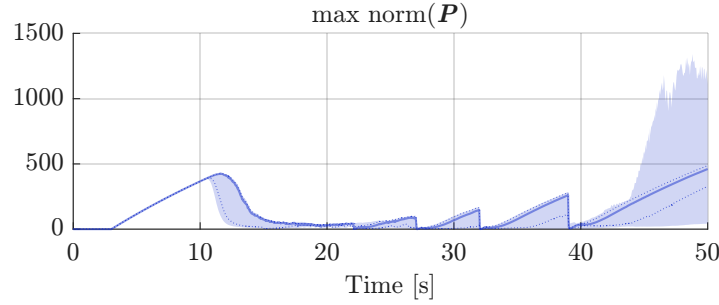


Figure 6.10: Covariance norm of the EKF in Monte-Carlo simulations. Solid line is the median value, dotted lines mark the 80th percentile, transparent fill includes 95% of all simulations.

6.4 Hardware-in-the-loop

Hardware-in-the-loop (HIL) testing was performed to validate the embedded implementation of the controller and estimator as a whole. Verification of individual model functions and algorithms was accomplished with unit tests beforehand. The HIL simulations are conducted with the flight processor running the embedded C code [LiEtAl25]. A computer running the CLOSEDROCKET simulation is connected to the processor over USB, and data is transmitted using UART. The simulation provides the processor with emulated sensor data over UART, while the processor feeds back servo commands to be used in the simulation, and transmits the estimated state for logging. The onboard RTOS of the processor is modified to use the simulation time provided by SIMULINK, such that processor and simulation advance in step.

Notably the UART connection between the processor and the computer is imperfect. Due to the specific implementation of the data synchronization, the simulation does not wait until transmissions from the processor are completely received nor does it acknowledge the reception of packets, causing some data to be delayed or lost. Additionally, even though both the processor and simulation can run in real time, the UART interface imposes a bottleneck on the transmission speed requiring the execution time to be slowed by a factor of 14. Further details on the implementation of the HIL framework exceed the scope of this thesis.

The simulated state is shown in Fig. 6.11, the parameters are similar to the nominal simulation (Sect. 6.2.1). Liftoff occurs at $t = 2$ s and apogee is reached at about $t = 53$ s. The controlled roll behavior is as expected very similar to the roll behavior of the nominal simulation, indicating that there are no major errors in the implementation.

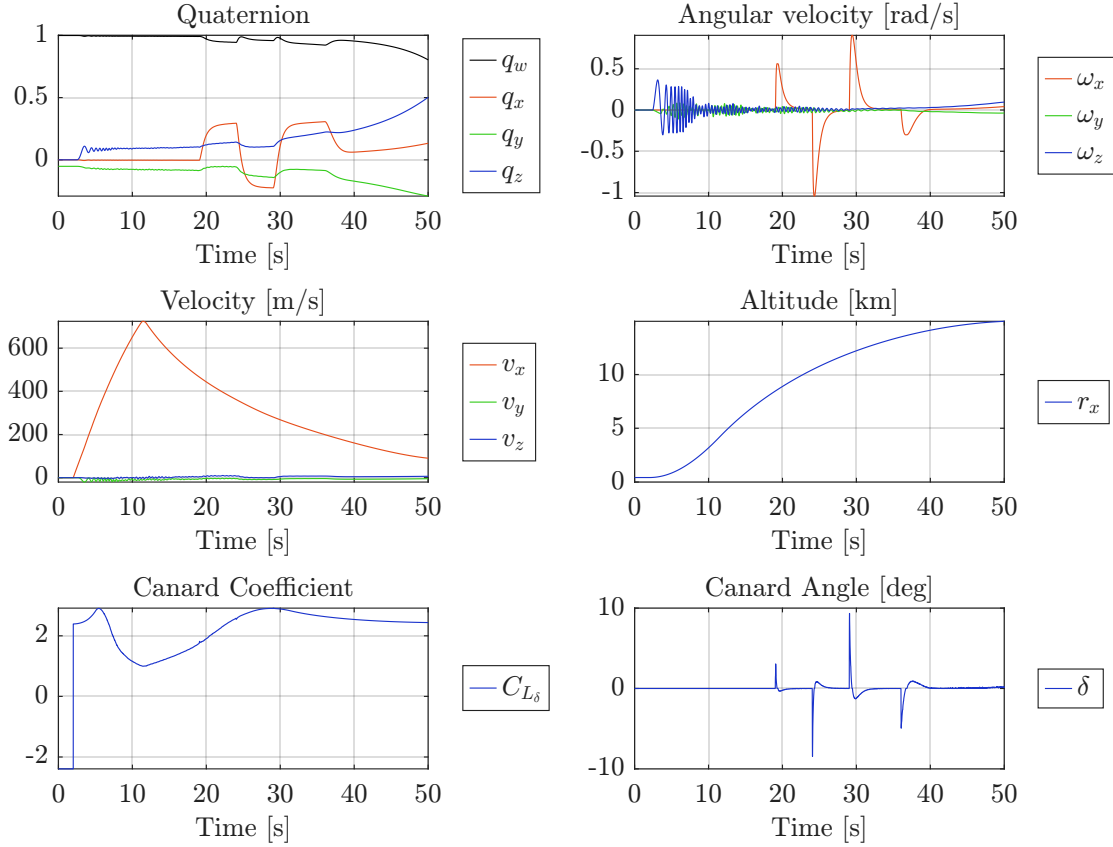


Figure 6.11: Simulated state of the HIL test. Nominal simulation parameters.

The estimation error is shown in Fig. 6.12. The data delays result in higher estimation errors during fast transients, as indicated by the thin spikes in the angular velocity and canard angle plots at the roll program steps $t = 17$ s, $t = 24$ s,

$t = 28$ s, and $t = 36$ s. Most notable is the large spike of the errors in pitch rate, velocity, and altitude at $t = 7$ s. Apart from these spikes the estimation error magnitudes are comparable to those of the nominal simulation. Furthermore, as the errors quickly recover after a spike it is likely that these were caused by individual transmission errors. Such a transmission error could occur in either direction of the interface, such that a false value of a sensor emulation, e.g. acceleration or barometric pressure, could cause the large estimation error at $t = 7$ s.

A mathematical error is another possible cause for the estimation error spikes, but unlikely due to broad unit test coverage. It is therefore reasonable to conclude that the controller and estimator perform as intended. However, the UART interface should be improved for future HIL testing with respect to interface speed and reliability.

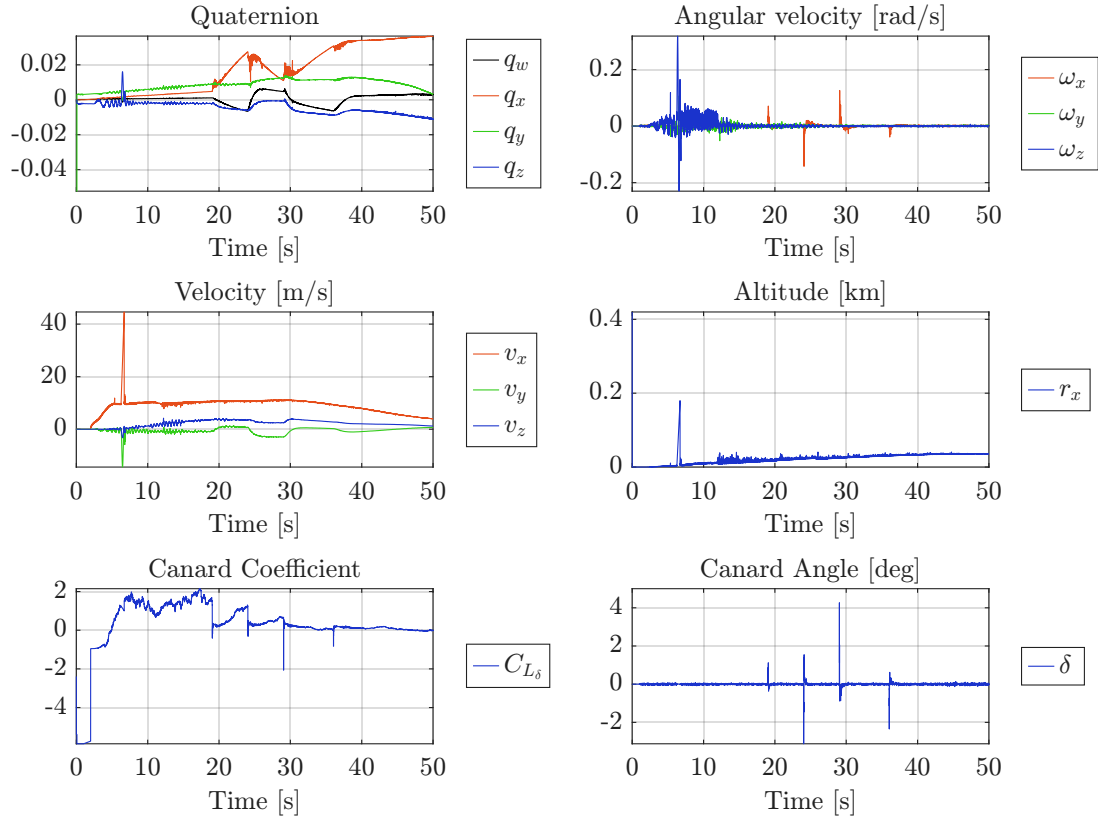


Figure 6.12: Estimation error of the HIL test. Nominal simulation parameters.

Chapter 7

Flight results

A test flight was conducted to validate the hard- and software under flight conditions, the results are presented in Sect. 7.1. The ultimate test is the flight of Aurora at Launch Canada 2025, the results are analyzed in Sect. 7.2.

7.1 Testflight results

The test flight was conducted successfully on May 10th 2025 near Louisville, Kentucky. Due to an error with internal logging only the telemetry data is available. This data was transmitted by the processor via the rocket CAN bus and sent over radio telemetry to the ground station computer, where it was finally logged. The resulting state estimation data is shown in Fig. 7.1, including the attitude as Euler angles for clarity.

The rocket lifts off at $t = 5$ s and the boost phase lasts for about 3 s. Following the unpowered ascent, apogee was reached at about $t = 25$ s. The test flight rocket is much lighter and shorter than Aurora, and used a solid rocket motor with significantly less impulse than Aurora's engine. This resulted in a shorter, slower, and lower flight profile compared to Aurora. At $t = 27$ s the drogue parachute was deployed, at which point the primary control mission ends. Similar to simulations the deployment caused a period of erratic tumbling as the rocket is pulled around by the parachute.

Due to human error and unaccounted differences between the launch operations of the test flight and Aurora, a safety lockout in software was triggered by a timer several seconds before liftoff. This caused the controller to be deactivated and the canards to be locked to a neutral position, such that the rocket had no control authority for the entirety of flight. Slight fin cant or some misalignment of the canards led to a slow roll rate ω_x with a peak of about 2 rad/s as indicated in the angular velocity plot. As the controller was inactive during the flight the canard angle δ is zero everywhere. Furthermore, the measurement feedback for

the coefficient C_L was deactivated for this test flight and thus the coefficient is constant. This choice was made because Monte-Carlo simulations were not yet conducted, and the coefficient estimation was deemed unnecessary for the test flight due to less extreme aerodynamics compared to Aurora.

Overall the estimated state looks as expected for an uncontrolled flight. The quaternion attitude and angular velocity are similar to the simulation in Fig. 6.4, albeit significantly slower. The dynamic pressure of the test flight was only about one tenth of the dynamic pressure of the Aurora simulation, and the boost phase lasted for only 3 s compared to the 10 s boost of Aurora. The state estimator continued running successfully during parachute deployment and descent. When transforming the velocity from the body-fixed frame to the ground frame, the vertical velocity estimate and the altitude estimate show the same descent speed of about -23 m/s. It is reasonable to conclude that the state estimator performed as intended.

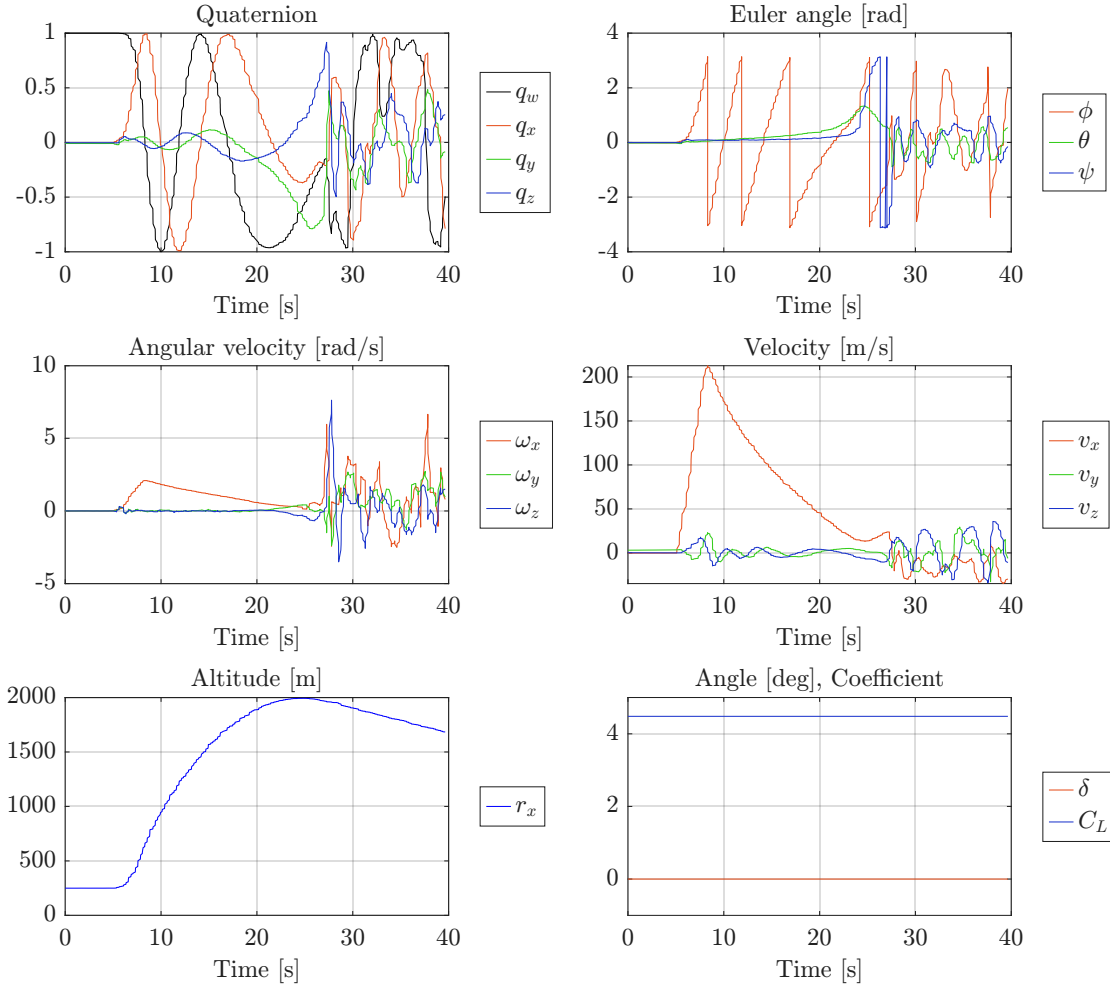


Figure 7.1: Test flight telemetry. Estimated state.

7.2 Aurora flight results

Aurora launched on August 21st 2025 at the Launch Canada Challenge [LC25] near Timmins, Ontario. Telemetry was received in flight from the controls processor and other onboard computers. This data indicates deployment of the recovery system, however no parachute deployment was observed due to cloud cover, and the descent rate points to a ballistic trajectory. Efforts to recover the rocket were unsuccessful, and Aurora remains lost.

Telemetry data includes the estimated state, controller commands, and measurements from onboard sensors. In the following plots, liftoff occurs at $t = 2$ s, the boost phase ends at $t = 12$ s, and apogee is reached at $t = 52$ s. Dots are used to indicate a received data point, they are connected for clarity.

The commands and sensor data are shown in Fig. 7.2. As described in Sect. 6.2.3, competition rules dictate that the controller shall be deactivated during the boost phase. This is implemented by a 10 s timer which is started at liftoff, so commands start at $t = 12$ s in this plot. Many data points of the commands were received, so the dots are omitted. The controller commands oscillate strongly during flight, much more than expected from pre-flight simulations. This might be caused by a much lower control authority than expected, or due to a misalignment between the canard and the encoder. Encoder data indicates the latter, as the encoder angle never reaches 10 deg on one side, and starts with small offset of about 1° .

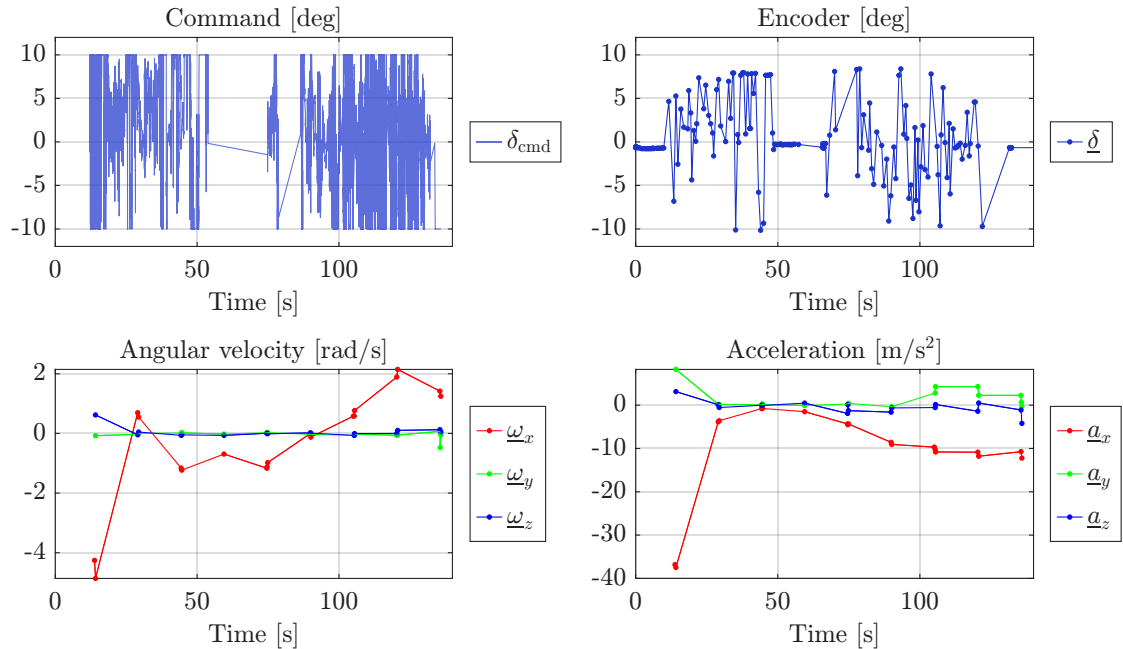


Figure 7.2: Aurora telemetry. Controller, encoder, and sensor data.

The most probable explanations are either that the alignment process was not accurate, or that the encoder set screw came loose at some point in assembly and the canard shaft slipped in the encoder hub.

The estimated state is shown in Fig. 7.3, including the attitude in Euler angles for clarity. The altitude estimate shows an apogee of 12.24 km above sea level, which is within 200 m of the recorded apogee from the commercially available altimeters. The maximum velocity estimate indicates a top speed of 638 m/s, which corresponds to about twice the speed of sound at that altitude. As only a few data points are available at the end of the boost phase, it is likely that the top speed was slightly higher.

The first in-flight data points of the attitude, angular velocity, and velocity estimates indicate a significant roll spin-up during the boost phase. Angular velocity measurements from the gyroscope also indicate a similar roll rate. Notably, the roll rate ω_x was roughly -10 rad/s at $t = 10$ s, two seconds before the rocket is under maximum dynamic pressure. The controller was deactivated during the boost phase and a roll rate response of similar magnitude was expected, as described in Sect. 6.2.3. High lateral velocities of about $v_y = 170$ m/s and $v_z = 109$ m/s during the boost phase indicate either strong wind gusts or a significant coning motion. Inaccurate estimation cannot be ruled out however, as lateral velocities of this magnitude were not observed in simulations, as shown in Fig. 6.4.

The next angular velocity data points are available a few seconds later. Both gyroscope measurements and state estimates show a significant reduction in the roll rate. At $t = 14$ s the roll rate is -5 rad/s, and six seconds later at $t = 20$ s the roll rate is reduced to -0.12 rad/s. The roll rate does not exceed -1.25 rad/s for the remainder of the ascent. Furthermore, ten complete quaternion data points were received between $t = 20$ s and $t = 52$ s, all showing a roll angle ϕ between -0.3 rad/s and 1.3 rad/s. The reasonable conclusion is that the controller successfully reduced the roll rate by a significant amount. There are not enough data points to analyze the response to the roll program.

The estimated canard coefficient of lift shows a large increase during the boost phase. This could be due to the roll spin-up being caused by disturbances while the canards were locked straight, effectively estimating a high effectiveness of the small remaining canard angle. After the controller is activated the coefficient quickly reduces to the expected order of magnitude, staying between 0.1 and 0.8 for the rest of the ascent. At about $t = 60$ s the coefficient estimate is again large, this corresponds to the time of low observability due to the low velocity.

During the entire descent both attitude and velocity indicate that the rocket fell with the forward end down. This supports the theory that the parachute was not deployed correctly, which was indicated by the fast descent rate.

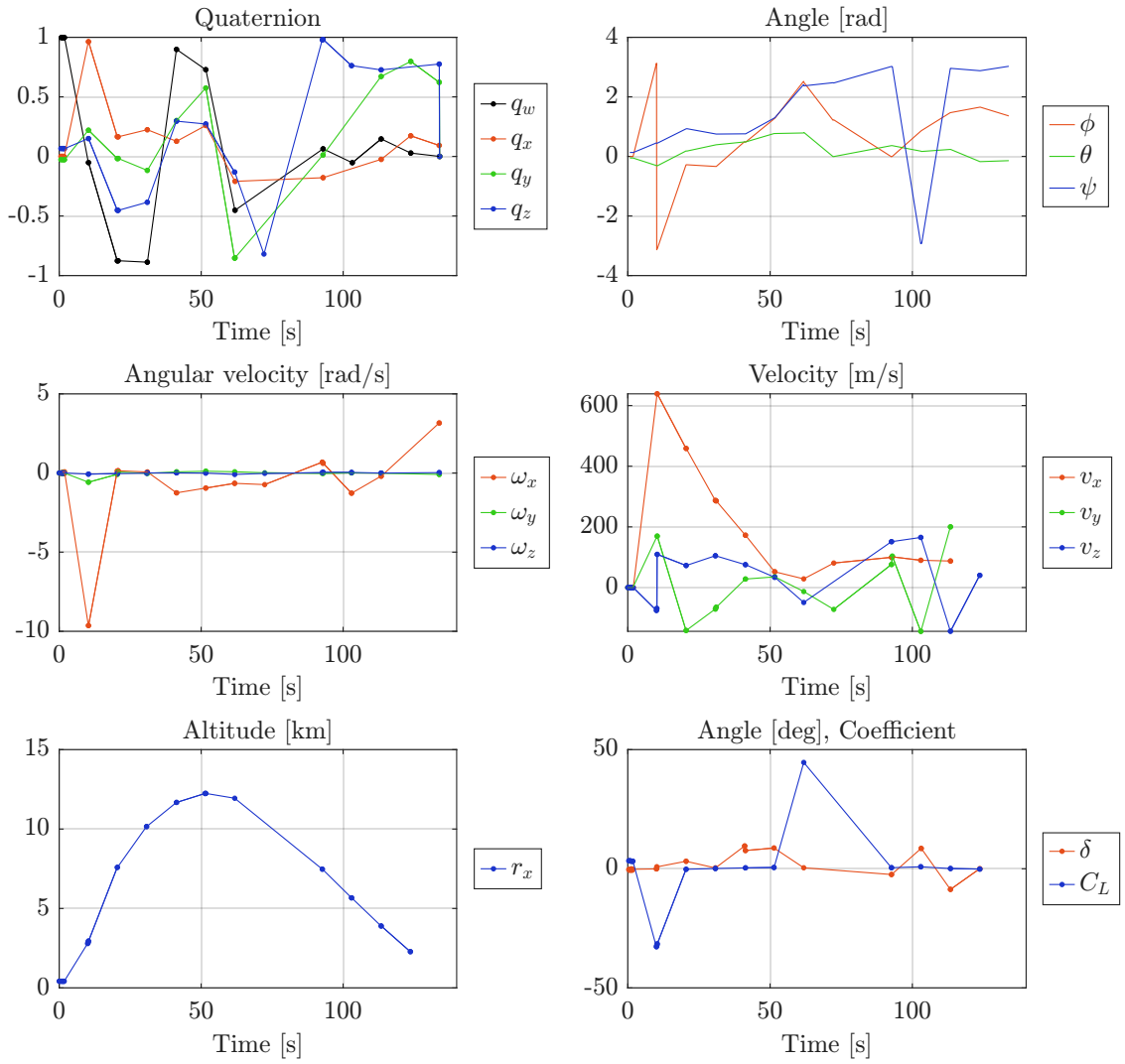


Figure 7.3: Aurora telemetry. Estimated state.

Chapter 8

Conclusion

8.1 Summary

This thesis developed a roll controller and state estimator for canard-controlled sounding rockets. The central problem addressed in this work was the mitigation of coning motion and prevention of roll lock-in during the ascent phase of the Aurora rocket, by using canard surfaces for active roll stabilization.

In Chap. 2, the problem of lock-in was described and the Aurora system was presented. While passive stabilization methods can reduce the chance of lock-in, they cannot guarantee prevention due to the chaotic nature of the phenomenon.

The control development was conducted in three parts. First, the flight dynamics were modeled in Chap. 3. Quaternions are used for the attitude parametrization, and models were developed for the onboard sensors and aerodynamics. Second, the roll controller was designed in Chap. 4. The proposed control scheme consists of LQR feedback and a feedforward roll program. Both state and flight condition for gain scheduling are provided by the estimator. When accounting for actuator delay, the robustness margins of the control loop are 14.5 dB in gain and 54° in phase. Third, the state estimator was developed in Chap. 5. The proposed EKF determines the rocket's attitude, rates, velocity, altitude, and canard parameters from redundant accelerometers, gyroscopes, magnetometers, barometers, and actuator encoders. A pad filter determines the initial state and sensor biases on the launch pad.

The control loop was analyzed by simulation in Chap. 6. Comparing a controlled with an uncontrolled flight, the controller successfully suppressed coning oscillations which were present without active control. Monte-Carlo simulations were used to analyze the robustness of the control loop to plant/model mismatch and disturbances. The controller was able to stabilize the roll rate in over 80% of simulations, even when roll reversal was present. Under roll reversal the estimator was able to detect the changed coefficient of lift of the canards and controller

gains were adjusted by the schedule. Roll rates grew when external disturbances were too large for the canard control authority. The estimation error was shown to be bounded for at least 95 % of simulations.

Two flights with the canard control system were conducted, and results were presented in Chap. 7. On the test flight the controller was deactivated, however the estimator was shown to work with reasonable certainty. Aurora was launched successfully, but was lost due to a recovery failure. Thus only limited telemetry data is available. Telemetry indicates that during ascent the roll was regulated from a rate of 10rad/s to below 1.25rad/s after the controller was activated. Additionally, state estimates match data from commercial flight computers.

The results of the test flight confirm that the developed estimation method is suitable for sounding rockets. Even though the limited flight data from Aurora cannot show conclusively that the controller worked, it still indicates successful roll control and supports further development.

8.2 Future work

Collecting conclusive flight data of a working roll control system may pave the way for a full attitude control system in the future. Pitch/yaw stability augmentation should be able to reduce the recovery dispersion and increase the apogee of sounding rockets. For full attitude control, direct quaternion feedback could be used instead of the projection to Euler angles [SongEtAl01], which could solve chattering problems around Euler angle discontinuities [MayhewSanfeliceTeel11].

Due to the gain scheduling approach the controller depends on large pre-computed gain tables. Instead, solving the LQR problem online could decrease the file sizes and improve adaptability. Closed-form solutions of some LQR problems exist [StevensJohnsonLewis15], such that no solvers would be required for a simpler roll model. For improved tracking performance the reference gain could be replaced with a dynamic feedforward controller, as shown in [TheisEtAl15].

The estimation accuracy depends strongly on the fidelity of the models. For high-altitude flights the constant gravity vector could be replaced by an altitude dependent gravity model, such as WGS84-EGM [StevensJohnsonLewis15]. The explicit Euler integration is simple to implement, but suffers in accuracy. Especially the attitude and velocity estimates would benefit from either more accurate numerical methods as shown in [SveierSjøbergEgeland19], or a closed-form integration as shown in [Solà17]. Additionally, the EKF structure could be modified to comply with the quaternion geometry during correction feedback and the covariance process, as shown with the invariant EKF [BonnabelMartinSalaun09] or the discrete Lie EKF [BourmaudEtAl13].

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Appendix A

Contents Archive

There is a folder **PRO_089_Breuer/** in the archive. The main folder contains the entries

- **PRO_089_Breuer.pdf**: the pdf-file of the thesis PRO-089.
- **Data/**: a folder with all the relevant data, programs, scripts and simulation environments.
- **Latex/**: a folder with the *.tex documents of the thesis PRO-089 written in Latex and all figures (also in *.svg data format if available).
- **Presentation/**: a folder with the relevant data for the presentation including the presentation itself, figures and videos.

Appendix B

Jacobians

In this appendix, Jacobians of the model (Chap. 3) will be derived analytically.

The Jacobian $\mathbf{J}_x$ is the matrix of all first-order partial derivatives of a vector-valued function. For the prediction and measurement model, the Jacobians of the functions $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are needed with respect to the state $\mathbf{x} \in \mathbb{R}^n$. The Jacobian of any function with respect to $\mathbf{x}$ is defined as

$$\mathbf{J}_x = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1} & \dots & \frac{\partial h_m}{\partial x_n} \end{bmatrix}. \quad (\text{B.1})$$

Partial derivatives of a function are here denoted as $X_Y = \frac{\partial}{\partial Y} X(Y)$. Note that elements in vectors and matrices are indexed by w, x, y, z , these do not denote a partial derivative. The dynamics and measurement model Jacobians are pieced together from the Jacobians of the individual functions. The variable X is the output row, and the index $_Y$ denotes the (state) column.

The continuous-time dynamics Jacobian $\mathbf{F} \in \mathbb{R}^{n \times n}$ is

$$\mathbf{F} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \dot{\mathbf{q}}_q & \dot{\mathbf{q}}_\omega & 0 & 0 & 0 & 0 \\ 0 & \dot{\boldsymbol{\omega}}_\omega & \dot{\boldsymbol{\omega}}_v & 0 & \dot{\boldsymbol{\omega}}_{C_L} & \dot{\boldsymbol{\omega}}_\delta \\ \dot{\mathbf{v}}_q & \dot{\mathbf{v}}_\omega & \dot{\mathbf{v}}_v & 0 & 0 & 0 \\ \dot{r}_{x,q} & 0 & \dot{r}_{x,v} & 0 & 0 & 0 \\ 0 & 0 & \dot{C}_{L,v} & 0 & -\alpha_c & 0 \\ 0 & 0 & 0 & 0 & 0 & -\alpha_\delta \end{bmatrix}. \quad (\text{B.2})$$

Integration to discrete-time is done with explicit Euler, which results in the discrete-time dynamics Jacobian

$$\mathbf{F}_D = \frac{\partial \mathbf{f}_D}{\partial \mathbf{x}} = \mathbf{I} + dt \mathbf{F}. \quad (\text{B.3})$$

The combined measurement Jacobian $\mathbf{H} \in \mathbb{R}^{m \times n}$ is

$$\mathbf{H} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} 0 & \mathbf{I}_3 & 0 & 0 & 0 & 0 \\ \mathbf{m}_q & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & p_{r_x} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (\text{B.4})$$

B.1 Kinematics and Quaternions

The cross product Jacobian is the skew symmetric matrix

$$(\mathbf{u} \times \mathbf{v})_{\mathbf{u}} = -\tilde{\mathbf{v}}, \quad (\mathbf{u} \times \mathbf{v})_{\mathbf{v}} = \tilde{\mathbf{u}}, \quad (\text{B.5})$$

with the tilde operator

$$\tilde{\mathbf{u}} = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}. \quad (\text{B.6})$$

B.1.1 Quaternion dynamics

Recall the Quaternion multiplication from Eq. (3.7)

$$\mathbf{p} = \mathbf{q} * \mathbf{q} = \tilde{\mathbf{Q}} \mathbf{q} \quad \tilde{\mathbf{Q}} = \begin{bmatrix} q_w & -q_x & -q_y & -q_z \\ q_x & q_w & -q_z & q_y \\ q_y & q_z & q_w & -q_x \\ q_z & -q_y & q_x & q_w \end{bmatrix}, \quad (\text{B.7})$$

and the time derivative from Eq. (3.8)

$$\dot{\mathbf{q}} = \frac{1}{2} \tilde{\mathbf{Q}} \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} = \frac{1}{2} \tilde{\boldsymbol{\Omega}} \mathbf{q} \quad \text{with} \quad \tilde{\boldsymbol{\Omega}} = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}. \quad (\text{B.8})$$

The Jacobian matrices for the quaternion time derivative are

$$\dot{\mathbf{q}}_{\mathbf{q}} = \frac{1}{2} \tilde{\boldsymbol{\Omega}} = \frac{1}{2} \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}, \quad (\text{B.9})$$

$$\dot{\mathbf{q}}_{\boldsymbol{\omega}} = \frac{1}{2} \tilde{\mathbf{Q}}_{:,2:4} = \frac{1}{2} \begin{bmatrix} -q_x & -q_y & -q_z \\ q_w & -q_z & q_y \\ q_z & q_w & -q_x \\ -q_y & q_x & q_w \end{bmatrix}. \quad (\text{B.10})$$

B.1.2 Quaternion rotations

The inverse of a unit quaternion is its conjugate [StevensJohnsonLewis15, Solà17]

$$\text{for } \|\mathbf{q}\| = 1 : \quad \mathbf{q}^{-1} = \begin{bmatrix} q_w \\ \mathbf{q}_v \end{bmatrix}^{-1} = \begin{bmatrix} q_w \\ -\mathbf{q}_v \end{bmatrix}. \quad (\text{B.11})$$

The rotation $\mathbf{S}(\mathbf{q}) \mathbf{v}$ is equivalent to the quaternion operation [Solà17]

$$\mathbf{R}(\mathbf{q}, \mathbf{v}) = \mathbf{S}(\mathbf{q}) \mathbf{v} \quad \Longleftrightarrow \quad \begin{bmatrix} 0 \\ \mathbf{R}(\mathbf{q}, \mathbf{v}) \end{bmatrix} = \mathbf{q}^{-1} * \begin{bmatrix} 0 \\ \mathbf{v} \end{bmatrix} * \mathbf{q} \quad (\text{B.12})$$

with the rotation matrix Eq. (3.9)

$$\begin{aligned} \mathbf{S} &= 2\mathbf{q}_v \mathbf{q}_v^\top + (q_w^2 - \mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3 - 2q_w \tilde{\mathbf{q}}_v \\ &= \begin{bmatrix} q_w^2 + q_x^2 - q_y^2 - q_z^2 & 2q_w q_z + 2q_x q_y & 2q_x q_z - 2q_w q_y \\ 2q_x q_y - 2q_w q_z & q_w^2 - q_x^2 + q_y^2 - q_z^2 & 2q_w q_x + 2q_y q_z \\ 2q_w q_y + 2q_x q_z & 2q_y q_z - 2q_w q_x & q_w^2 - q_x^2 - q_y^2 + q_z^2 \end{bmatrix}, \end{aligned} \quad (\text{B.13})$$

and $\mathbf{S}^\top = \mathbf{S}(\mathbf{q}^{-1})$. The Jacobian of rotation after the vector $\mathbf{u}$ is

$$\mathbf{R}_u(\mathbf{q}) = \mathbf{S}, \quad (\text{B.14})$$

and the Jacobian of rotation after the quaternion $\mathbf{q}$ is

$$\begin{aligned} \mathbf{R}_q(\mathbf{q}, \mathbf{u}) &= 2 \begin{bmatrix} q_w \mathbf{u} - \mathbf{q}_v \times \mathbf{u} & \mathbf{q}_v^\top \mathbf{u} \mathbf{I}_3 + \mathbf{q}_v \mathbf{u}^\top - \mathbf{u} \mathbf{q}_v^\top + q_w \tilde{\mathbf{u}} \end{bmatrix} = 2 \cdot \\ &\begin{bmatrix} q_w u_x - q_y u_z + q_z u_y & q_x u_x + q_y u_y + q_z u_z & q_x u_y - q_w u_z - q_y u_x & q_w u_y + q_x u_z - q_z u_x \\ q_w u_y + q_x u_z - q_z u_x & q_w u_z - q_x u_y + q_y u_x & q_x u_x + q_y u_y + q_z u_z & q_y u_z - q_w u_x - q_z u_y \\ q_w u_z - q_x u_y + q_y u_x & q_z u_x - q_x u_z - q_w u_y & q_w u_x - q_y u_z + q_z u_y & q_x u_x + q_y u_y + q_z u_z \end{bmatrix}. \end{aligned} \quad (\text{B.15})$$

B.1.3 Relative motion

The position change in the inertial frame from Eq. (3.11) results in the Jacobians

$$\dot{\mathbf{r}}_q = \mathbf{R}_q(\mathbf{q}^{-1}, \mathbf{v}), \quad (\text{B.16})$$

$$\dot{\mathbf{r}}_v = \mathbf{S}^\top. \quad (\text{B.17})$$

Only the altitude, and not the full spatial position, is used as a state. Thus just the first row of these matrices is used for r_x .

The velocity change in the body-fixed frame from Eq. (3.12) results in the Jacobians

$$\dot{\mathbf{v}}_q = \mathbf{R}_q(\mathbf{q}, \mathbf{g}^G), \quad (\text{B.18})$$

$$\dot{\mathbf{v}}_\omega = \tilde{\mathbf{v}}, \quad (\text{B.19})$$

$$\dot{\mathbf{v}}_v = -\tilde{\omega}, \quad (\text{B.20})$$

When the torque $\mathcal{T}$ is assumed to be approximately constant w.r.t $\rho(r_x)$, the angular rate change in the body-fixed frame from Eq. (3.13) results in the Jacobians

$$\dot{\omega}_\omega = -\mathbf{J}^{-1}(\tilde{\mathbf{J}}\omega), \quad (\text{B.21})$$

$$\dot{\omega}_v = \mathbf{J}^{-1}\mathcal{T}_v, \quad (\text{B.22})$$

$$\dot{\omega}_{C_L} = \mathbf{J}^{-1}\mathcal{T}_{C_L}, \quad (\text{B.23})$$

$$\dot{\omega}_\delta = \mathbf{J}^{-1}\mathcal{T}_\delta, \quad (\text{B.24})$$

where the torque Jacobians are derived in a later section. Note that the tilde operator is performed here on the product $\mathbf{J}\omega$.

B.2 Sensor measurements

The sensor data is assumed to be already rotated into the body-fixed frame. The acceleration enters the filter via the dynamics input, and is thus not part of the formal measurement prediction.

The Jacobian of the gyroscope measurement is

$$\underline{\omega}_\omega = \mathbf{I}_3, \quad (\text{B.25})$$

and the Jacobian of the magnetic field measurement is

$$\underline{\mathbf{m}}_q = \mathbf{R}_q(\mathbf{q}, \mathbf{m}_E^G). \quad (\text{B.26})$$

The pressure measurement has the Jacobian (atmosphere partial derivatives are treated in the next section)

$$\underline{p}_{r_x} = p_{r_x}, \quad (\text{B.27})$$

and the canard angle encoder has the trivial Jacobian

$$\underline{\delta}_\delta = 1. \quad (\text{B.28})$$

B.3 Atmosphere and air data

Partial derivatives of the atmosphere model from Eq. (3.20) to Eq. (3.24) after the geographic altitude r_x are

$$\ell_{r_x} = \left(\frac{R_0}{R_0 - r_x} \right)^2, \quad (\text{B.29})$$

$$T_{r_x} = -k_b \ell_{r_x}, \quad (\text{B.30})$$

$$p_{r_x} = \begin{cases} p_b \ell_{r_x} \frac{g_0}{RT_b} \exp\left(\frac{-g_0}{RT_b}(r_x - \ell_b)\right), & \text{for } k_b \neq 0 \\ p_b \ell_{r_x} \frac{g_0}{RT_b} \left(1 - \frac{k_b}{T_b}(\ell - \ell_b)\right)^{\frac{g_0}{Rk_b} - 1}, & \text{for } k_b = 0 \end{cases} \quad (\text{B.31})$$

$$\rho_{r_x} = \frac{p_{r_x} T - p T_{r_x}}{R T^2}, \quad (\text{B.32})$$

$$c_{r_x} = \frac{1}{2} T_{r_x} \sqrt{\frac{\gamma R}{T}}. \quad (\text{B.33})$$

A the model depends on the atmosphere layers from Tab. 3.1, the derivatives are also treated piecewise with their respective base values.

B.4 Aerodynamics

The dynamics require the partial derivatives of the external torque $\mathcal{T}$ from Eq. (3.29) and Eq. (3.31)

$$\mathcal{T} = \bar{p} \left(A_{\text{can}} d_{\text{can}} C_{L_\delta} \begin{bmatrix} \delta \\ 0 \\ 0 \end{bmatrix} + A_{\text{ref}} d_{\text{cp}} C_{N_\alpha} \begin{bmatrix} 0 \\ \sin(\alpha) \\ -\sin(\beta) \end{bmatrix} \right). \quad (\text{B.34})$$

The density variation w.r.t the altitude is assumed to be very small, and therefore all $\mathcal{T}_{r_x} \approx \mathbf{0}$.

Because the rocket is stable in the pitch and yaw direction, the incidences can be linearized

$$v_x > 0 : \quad \alpha \approx v_z/v_x \quad \beta \approx v_y/v_x, \quad (\text{B.35})$$

$$v_x \leq 0 : \quad \alpha \approx \pi/2 \quad \beta \approx \pi/2. \quad (\text{B.36})$$

Partial derivatives of the dynamic pressure $\bar{p}$, and the product of pressure and incidence angles are

$$\bar{\mathbf{p}}_v = \rho \mathbf{v}^\top, \quad (\text{B.37})$$

$$\left(\bar{p} \begin{bmatrix} \sin(\alpha) \\ \sin(\beta) \end{bmatrix} \right)_v \approx \frac{\rho}{2} \begin{bmatrix} v_z & 0 & v_x + 3v_z^2/v_x \\ v_y & v_x + 3v_y^2/v_x & 0 \end{bmatrix} \quad (\text{B.38})$$

$$\approx \frac{\rho}{2} \begin{bmatrix} v_z & 0 & v_x \\ v_y & v_x & 0 \end{bmatrix}. \quad (\text{B.39})$$

Thus Jacobians of the external torques are

$$\mathcal{T}_v = \rho A_{\text{can}} d_{\text{can}} C_{L\delta} \begin{bmatrix} \delta \\ 0 \\ 0 \end{bmatrix} \mathbf{v}^\top \quad (\text{B.40})$$

$$+ \frac{\rho}{2} A_{\text{ref}} d_{\text{cp}} C_{N\alpha} \begin{bmatrix} 0 & 0 & 0 \\ v_z & 0 & v_x \\ -v_y & -v_x & 0 \end{bmatrix}, \quad (\text{B.41})$$

$$\mathcal{T}_{C_L} = \bar{p} A_{\text{can}} d_{\text{can}} \begin{bmatrix} \delta \\ 0 \\ 0 \end{bmatrix}, \quad (\text{B.42})$$

$$\mathcal{T}_\delta = \bar{p} A_{\text{can}} d_{\text{can}} C_{L\delta} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \quad (\text{B.43})$$

B.5 Airfoil and actuator

The partial derivatives of the canard airfoil are treated in Sect. C.1.

The partial derivative of the actuator dynamics is simply

$$\dot{\delta}_\delta = -\alpha_\delta. \quad (\text{B.44})$$

Appendix C

Further models

C.1 Canard airfoil theory

In supersonic flow protruding surfaces of the rocket body, such as the canards, produce shock waves. At higher Mach numbers the shock cone becomes steeper, changing the flow behavior and thus the lift coefficient. The characteristic Mach angle is defined as $\mu = 0$ for $\text{Ma} < 1$, and for $\text{Ma} \geq 1$ as

$$\mu = \arccos\left(\frac{1}{\text{Ma}}\right). \quad (\text{C.1})$$

Aurora’s canards have a delta planform as shown in Fig. C.1, with a high sweep angle of $\Lambda = 61^\circ$.

The coefficient of lift for delta wings can be approximated by models given in Stengel [Stengel04]. For supersonic airspeeds and a subsonic leading edge $\mu \leq \Lambda$, where the canard surface is completely behind the shock cone, the coefficient of lift is determined by an empirical function. Both for subsonic airspeeds $\text{Ma} < 1$ and for supersonic leading edges $\mu > \Lambda$ simpler laws suffice. The coefficient of

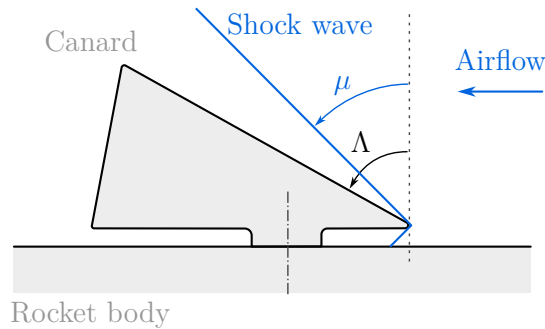


Figure C.1: Diagram of the canard planform and tip shock cone.

lift derivative is thus defined piecewise as [Stengel04]

$$\text{Ma} < 1 : \quad C_{L_\delta} = 2\pi \cot(\Lambda) , \quad (\text{C.2})$$

$$\mu < \Lambda : \quad C_{L_\delta} = \frac{2\pi^2 \cot(\Lambda)}{\pi + \eta(0.38 + 2.26\eta - 0.86\eta^2)} \quad \text{with} \quad \eta = \frac{\cot(\Lambda)}{\cot(\mu)} , \quad (\text{C.3})$$

$$\mu \geq \Lambda : \quad C_{L_\delta} = \frac{4}{\sqrt{\text{Ma}^2 - 1}} . \quad (\text{C.4})$$

This coefficient for different Mach numbers is shown in Fig. C.2, the linear model is derived next.

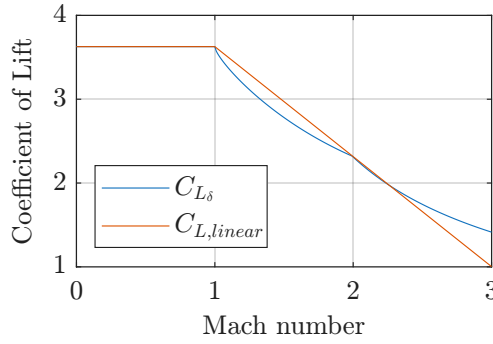


Figure C.2: Canard coefficient of lift approximation.

As the theoretical value is cumbersome to partially differentiate, it is practical to assume it is linear w.r.t. all variables. The lift curve intersects the points $C_{L_\delta}(\text{Ma} = 1)$, and $C_{L_\delta}(\mu = \Lambda)$. The latter is the supercritical point, at the Mach number $\text{Ma}_{\text{sc}} = \frac{1}{\cos(\Lambda)}$. The linear approximation is then

$$\text{Ma} \leq 1 : \quad C_{L,\text{linear}} = 2\pi \cot(\Lambda) , \quad (\text{C.5})$$

$$\text{Ma} > 1 : \quad C_{L,\text{linear}} = 2\pi \cot(\Lambda) + \underbrace{\frac{(2\pi - 4.01) \cot(\Lambda)}{1 - 1/\cos(\Lambda)}}_{C_{L,\text{slope}}} (\text{Ma}_{\text{sc}} - 1) , \quad (\text{C.6})$$

which is shown in Fig. C.2. The constant factor $C_{L,\text{slope}}$ can be pre-computed. This results in the following linearized partial derivatives, including the first order convergence model from Eq. (5.16)

$$\dot{C}_{LC_L} = -\alpha_c , \quad (\text{C.7})$$

$$\dot{C}_{Lv} = 0 \quad \text{for} \quad \text{Ma} \leq 1 , \quad (\text{C.8})$$

$$\dot{C}_{Lv} = \frac{\alpha_c}{c} C_{L,\text{slope}} \quad \text{for} \quad \text{Ma} > 1 , \quad (\text{C.9})$$

$$\dot{C}_{Lr_x} \approx 0 . \quad (\text{C.10})$$

C.2 Pad filter inclinometer

On the launch pad the accelerometer measures the gravitational acceleration, which is used to determine the initial orientation of the rocket. The vectors are normed such that $\|\mathbf{a}\| = 1 = \|\mathbf{g}\|$. As the normed gravity is $\mathbf{g}^G = \begin{bmatrix} -1 & 0 & 0 \end{bmatrix}^\top$, the measured acceleration is

$$\mathbf{a} = -\mathbf{S}(\mathbf{q}_0) \mathbf{g}^G = \begin{bmatrix} q_w^2 + q_x^2 - q_y^2 - q_z^2 \\ 2q_xq_y - 2q_wq_z \\ 2q_wq_y + 2q_xq_z \end{bmatrix}, \quad (\text{C.11})$$

with the quaternion norm $\|\mathbf{q}\| = 1$, so $q_w^2 + q_x^2 - q_y^2 - q_z^2 = 1 - 2(q_y^2 + q_z^2)$. Because the complete 3D orientation cannot be uniquely determined from a 3D vector measurement, the initial roll component is defined as zero

$$q_x = 0. \quad (\text{C.12})$$

Using this, Eq. (C.11) can be written as

$$q_w^2 = a_x + q_y^2 + q_z^2, \quad q_z = \frac{-a_y}{2q_w}, \quad q_y = \frac{a_z}{2q_w}. \quad (\text{C.13})$$

Rearranging while making use of the quaternion norm, and inserting q_y and q_z results in

$$a_x = 1 - 2(q_y^2 + q_z^2) = 1 - \frac{1}{2q_w^2} (a_y^2 + a_z^2), \quad (\text{C.14})$$

$$\implies \frac{1}{2q_w^2} = \frac{1 - a_x}{a_y^2 + a_z^2} \implies q_w = \sqrt{\frac{1}{2} \frac{a_y^2 + a_z^2}{1 - a_x}}. \quad (\text{C.15})$$

As the acceleration is normed, it follows that $a_y^2 + a_z^2 = 1 - a_x^2$ and therefore

$$q_w = \sqrt{\frac{1}{2} \cdot \frac{1 - a_x^2}{1 - a_x}} = \sqrt{\frac{1}{2} (1 + a_x)}. \quad (\text{C.16})$$

Inserting q_w into Eq. (C.13) results in the q_y and q_z components of the initial orientation.

C.3 Simplification of roll attitude dynamics

The time derivative of the roll angle Eq. (3.10)

$$\phi = \text{atan2} \left(2(q_w q_x + q_y q_z), (q_w^2 - q_x^2 - q_y^2 + q_z^2) \right) \quad (\text{C.17})$$

is

$$\begin{aligned} \dot{\phi} = & - \left(2(-\dot{q}_x q_w^3 + \dot{q}_w q_w^2 q_x - \dot{q}_z q_w^2 q_y - \dot{q}_y q_w^2 q_z - \dot{q}_x q_w q_x^2 \right. \\ & - 2\dot{q}_y q_w q_x q_y + 2\dot{q}_z q_w q_x q_z + \dot{q}_x q_w q_y^2 + 2\dot{q}_w q_w q_y q_z - \dot{q}_x q_w q_z^2 \\ & + \dot{q}_w q_x^3 + \dot{q}_z q_x^2 q_y + \dot{q}_y q_x^2 q_z + \dot{q}_w q_x q_y^2 - 2\dot{q}_x q_x q_y q_z \\ & - \dot{q}_w q_x q_z^2 + \dot{q}_z q_y^3 - \dot{q}_y q_y^2 q_z + \dot{q}_z q_y q_z^2 - \dot{q}_y q_z^3) \Big) \\ & / \left(q_w^4 + 2q_w^2 q_x^2 - 2q_w^2 q_y^2 + 2q_w^2 q_z^2 + 8q_w q_x q_y q_z \right. \\ & \left. + q_x^4 + 2q_x^2 q_y^2 - 2q_x^2 q_z^2 + q_y^4 + 2q_y^2 q_z^2 + q_z^4 \right) . \quad (\text{C.18}) \end{aligned}$$

Inserting the quaternion derivative Eq. (3.8)

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} -q_x \omega_x - q_y \omega_y - q_z \omega_z \\ q_w \omega_x + q_y \omega_z - q_z \omega_y \\ q_w \omega_y - q_x \omega_z + q_z \omega_x \\ q_w \omega_z + q_x \omega_y - q_y \omega_x \end{bmatrix}, \quad (\text{C.19})$$

and simplifying results

$$\begin{aligned} \dot{\phi} = & (q_x \omega_x + q_y \omega_y + q_z \omega_z)(q_w^2 q_x + 2q_w q_y q_z + q_x^3 + q_x q_y^2 - q_x q_z^2) \\ & / \left(q_w^4 + 2q_w^2 q_x^2 - 2q_w^2 q_y^2 + 2q_w^2 q_z^2 + 8q_w q_x q_y q_z \right. \\ & \left. + q_x^4 + 2q_x^2 q_y^2 - 2q_x^2 q_z^2 + q_y^4 + 2q_y^2 q_z^2 + q_z^4 \right) . \quad (\text{C.20}) \end{aligned}$$

Approximating

$$\omega_y, \omega_z \approx 0 \quad (\text{C.21})$$

thus the attitude derivative is

$$\dot{\mathbf{q}} \approx \frac{1}{2} \omega_x \begin{bmatrix} -q_x & q_w & q_z & -q_y \end{bmatrix}^\top, \quad (\text{C.22})$$

and reducing Eq. (C.20) results in

$$\dot{\phi} = \omega_x . \quad (\text{C.23})$$

Erklärung

Ich, Finn Jonas Breuer (Student des Theoretischen Maschinenbaus an der Technischen Universität Hamburg, Matrikelnummer 53744), versichere, dass ich die vorliegende Projektarbeit selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel verwendet habe. Die Arbeit wurde in dieser oder ähnlicher Form noch keiner Prüfungskommission vorgelegt.

Unterschrift

Datum